
Delayed Bandits with Drifting State Losses:

Effective-Dimension Pooling and Representation Tradeoffs

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Abstract

A learner chooses an action, sees an intermediate state at once, and learns the outcome only after a fixed delay. Because the outcome’s conditional mean depends on the action only through the state it reached, every action that might have reached that state shares in what the outcome reveals, so one delayed observation can update the whole action set rather than the action played. STATE-EXP3 does this and attains $\tilde{O}(\sqrt{(d+1)V \log K})$ for known action-to-state matrix P , where $V = \sum_t v_t$ and each *effective dimension* v_t lies in $[1, |\mathcal{S}|]$ and equals one plus the χ^2 mutual information between action and state. This replaces the action count by an overlap-sensitive dimension, improving on the $\tilde{O}(\sqrt{(K+d)T})$ available once the state-to-loss map is unrestricted whenever $\bar{v}(d+1) \log K \lesssim K+d$, when many actions are resolved by comparatively few shared states. Coarsening the states trades that dimension against the spread of losses inside a group, and unknown P is estimable from the immediate observations, though that guarantee is conservative. On the reported synthetic instances the practical $m=1$ variant, which our theorem does not cover, improves on action-level EXP3 by 5 to 22 per cent as K grows from 8 to 80. Delay has a second and separate cost. We prove a drift-dependent lower bound surviving even an exactly supplied stale action-loss vector, matching on single-action-drift instances, to logarithmic factors, the drift term of an existing upper bound. STATE-EXP3 does not adapt to that drift, so our bounds do not meet, and exploiting both channels at once is the problem this separation isolates.

1 Introduction

Delay is what makes many sequential decisions hard. A recommendation is served now and the purchase it was meant to cause is recorded weeks later, so a learner that waits learns slowly and one that does not has nothing to learn from. The usual response is to act on a signal that arrives at once, and the question is what it is worth.

Bandits with intermediate observations formalize this. An action produces a state at once and the loss attached to that state arrives after a fixed delay. Esposito et al. (2023) show the state-to-loss map decides the complexity and that the extremes are far apart. Over T rounds with K actions and delay d , a stationary map yields $\tilde{O}(\sqrt{(K + \min\{|\mathcal{S}|, d\})T})$ and an unrestricted one only the rate available with no intermediate observations at all.

The immediate state carries structural information both extremes leave unused. Observing s tells the learner its action reached s , and P says how likely every *other* action was to reach s , so one delayed outcome can update the entire action set in proportion to each action’s chance of having produced that state. We call this *pooling through the state*, and its price is neither the number of actions nor of states but an *effective dimension* measuring how much the rows of P overlap.

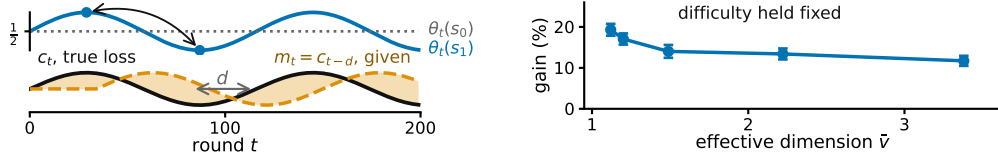


Figure 1: The two channels, both synthetic. **Left**, the temporal channel. θ_t moves while the sensor does not, so s_1 carries opposite losses at the marked rounds and the shaded area is the error E_2 accumulates. **Right**, the structural channel. With the uniform-play regret held approximately fixed, the gain of the practical $m = 1$ variant over action-level EXP3 rises as $\bar{v}$ falls, at $K = 40$, $|\mathcal{S}| = 8$, $T = 10000$, $d = 50$, 20 paired seeds, bars one standard error of the paired difference. Consistent with the mechanism of Theorem 1, which does not cover $m = 1$. Matching is approximate at the two smallest $\bar{v}$ (Appendix I, study 10).

Drift is a separate consequence of delay. Vernade et al. (2020) let the action-to-state map vary but hold the state-to-loss map fixed, the opposite of the component Esposito et al. (2023) identify as decisive, whereas applications sit between. A click predicts a purchase and an engagement signal predicts retention, relationships that drift slowly without turning adversarial (Figure 1). What a learner would want is the stale action-loss vector $m_t = c_t - d$, whose usefulness is governed by $E_2 = \sum_t \|c_t - m_t\|_\infty^2$. For this regime Bar-On and Mansour (2025) give a second-order upper bound but no lower bound.

Delay therefore creates two orthogonal bottlenecks and this paper characterizes each on its own. The *structural* channel is that immediate states reveal which actions share information, which makes pooling possible and which our algorithm exploits. The *temporal* channel is that stale predictions go out of date, which sets a floor on regret and which our lower bound prices. Nothing here exploits both, so a method pooling through states while adapting to drift is the problem this work isolates rather than solves.

- **Effective-dimension pooling.** STATE-EXP3 attains $\tilde{O}(\sqrt{(d+1)V \log K})$ for known P (Theorem 1, proved), and unknown P is estimable from the immediate observations, a warm-start version attaining $\tilde{O}(T^{4/5})$ (Appendix E, proved but conservative).
- **Representation trade-off.** The same algorithm on a fixed coarsening trades that dimension against the within-group spread of losses, making the state space a design choice (Theorem 2, proved).
- **The cost of drift.** A lower bound $\Omega(\sqrt{d E_2 \min\{1 + \log q, T/d\}})$ for q drifting coordinates, surviving the exact stale vector and gaining $\sqrt{KT}$ without it (Theorem 3, proved), the first for this model in E_2 to our knowledge. We conjecture the delay is charged additively and some algorithm adapts to E_2 , with two natural routes ruled out (Conjecture 4) and a supporting grid (Appendix I, empirical).

2 Related work

Optimal rates under arbitrary delays are due to Zimmert and Seldin (2020), at $\tilde{O}(\sqrt{KT} + \sqrt{D \log K})$ for total delay D . Bar-On and Mansour (2025) obtain $\tilde{O}(\sqrt{KT + \bar{\Lambda}})$ for a known inaccuracy budget, grading the accuracy of a changing observation where we grade the geometry of a fixed one, and Esposito et al. (2023) allow an unknown and possibly adversarial action-to-state map where ours is fixed and, for the upper bounds, known. Wang et al. (2026) give a lower bound in squared gradient prediction error with no delay, so the delay factor and its saturation are new here. Zhang et al. (2026) share the motivation, grading short-term indicators of a delayed outcome by an information measure and validating it in deployment, in a stochastic Bayesian model where ours is adversarial. The effective dimension itself is not new. Identifying an action with its distribution over outcomes makes $v_t - 1$ the χ^2 mutual information of Eldowa et al. (2024), whose policy-set capacity $\bar{v} - 1$ controls EXP4 regret under *mediator feedback*, where the outcome and its loss are both seen at once. Our contributions beyond this correspondence are the delayed and noisy analysis, the coarsening trade-off, and the drift lower bound. Appendix A adds optimism under delay and structural feedback sharing, where off-policy coverage (Gabbianelli et al., 2023) is the closest analogue.

76 3 Problem formulation

77 The base model comes first and the drift oracle afterwards, since only the lower bound uses it.
 78 Throughout, T is the number of rounds, K the number of actions, $\mathcal{S}$ a finite state space, d a fixed
 79 known delay, and all logarithms are natural.

80 Let $\mathcal{F}_{t-1}$ hold the outcomes of rounds $r \leq t - d - 1$, the states S_r for $r < t$ and the past ran-
 81 domization. At round $t \leq T$ the learner draws A_t from an $\mathcal{F}_{t-1}$ -measurable distribution x_t on
 82 $\{1, \dots, K\}$, observes $S_t \in \mathcal{S}$ drawn from $P(\cdot \mid A_t)$, and at time $t + d$ observes $X_t \in [0, 1]$ with
 83 $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}, A_t, S_t] = \theta_t(S_t)$, outcomes being conditionally independent across rounds given the
 84 states. Feedback generated at round r is revealed after the action at round $r + d$ has been selected
 85 and is first usable at round $r + d + 1$, so the learner acts on a d -round-old view of the losses.

86 The conditional mean $\theta_t(S_t)$ does not depend on A_t , so the state is sufficient for the outcome and an
 87 action affects the loss only through the state it reaches. This is what licenses charging one action's
 88 outcome to another, and an action acting by a route the state does not record breaks it.

89 Here $P \in [0, 1]^{K \times |\mathcal{S}|}$ is the *action-to-state matrix*, row-stochastic and fixed, and $\theta_t \in [0, 1]^{|\mathcal{S}|}$ is
 90 an oblivious sequence of *state-loss means*, so the *induced action-loss vector* $c_t = P\theta_t \in [0, 1]^K$
 91 averages the current state-loss means over the states each action reaches. Performance is the static
 92 pseudo-regret $\mathbb{E}[R_T] = \mathbb{E}[\sum_t c_t(A_t)] - \min_a \sum_t c_t(a)$, the expectation being over every source of
 93 randomness in the interaction, the learner's draws, the states and the outcomes. It is a *pseudo-regret*
 94 because it scores the conditional mean loss c_t rather than the realized outcome X_t .

95 For the lower bound only, we add a stronger *oracle model* in which the learner also receives the *stale*
 96 *action-loss vector* $m_t \triangleq c_{t-d}$ before choosing A_t , with $m_t = c_1$ for $t \leq d$. Under unknown P this
 97 is not a function of the observables, so a lower bound surviving it is stronger. STATE-EXP3 does
 98 not use m_t .

99 That model needs a measure of how misleading m_t is. Write $E_2 = \sum_t \|c_t - m_t\|_\infty^2 \leq T$, which
 100 squares the largest per-round error the stale vector makes and so stays small when errors are small
 101 though frequent. It is controlled by the state-loss variation W through $E_2 \leq d^2 W$, while no converse
 102 bound holds in general (Appendix G).

103 4 Pooling through the state

104 In this section and the next, P is known and the learner does not receive m_t . Because P records
 105 how likely every action was to reach the observed s , one outcome can be charged to all of them in
 106 proportion rather than spent on the action played.

107 Write $q_t(s) = \sum_a x_t(a)P(s \mid a)$ and set $\hat{c}_t(a) = P(S_t \mid a)X_t/q_t(S_t)$. This is unbi-
 108 ased for $c_t(a)$, and its play-weighted second moment is at most the *effective dimension* $v_t =$
 109 $\sum_s q_t(s)^{-1} \sum_a x_t(a)P(s \mid a)^2$, which lies in $[1, |\mathcal{S}|]$ (Appendix C) and replaces K below. Equiv-
 110 alently $v_t = 1 + \chi^2(P_{A_t, S_t} \parallel P_{A_t} \otimes P_{S_t})$, one plus the χ^2 mutual information between the action
 111 and the state it induces under x_t , so $v_t = 1$ exactly when the state carries no information about the
 112 action. STATE-EXP3 runs m interleaved copies of exponential weights, copy i taking the rounds
 113 congruent to i modulo m , playing $x_t \propto \exp(-\eta L_i)$ and adding $\hat{c}_t$ to L_i once X_t is usable, which
 114 for $m = d + 1$ is that copy's next round. The update has no optimization subproblem, costs $O(K)$
 115 given P and the scalar $q_r(S_r)$, and stores m weight vectors (Appendix C, with pseudocode).

116 **Theorem 1.** *Let P be known and (θ_t) oblivious. For every $\eta > 0$, STATE-EXP3 with $m = d + 1$
 117 satisfies $\mathbb{E}[R_T] \leq (d + 1) \log K / \eta + \frac{\eta}{2} \mathbb{E}[\sum_t v_t]$, so any V with $\sum_t v_t \leq V$ surely and $\eta =$
 118 $\sqrt{2(d + 1) \log K / V}$ give $\mathbb{E}[R_T] \leq \sqrt{2(d + 1)V \log K}$. Writing $\bar{v}$ for the largest v_t over all play
 119 distributions, $V = \bar{v}T \leq |\mathcal{S}|T$ is such a bound and depends on P alone.*

120 The effective dimension replaces the action count. It equals one when every action induces the same
 121 state distribution and reaches $|\mathcal{S}|$ when P is deterministic and onto, which still leaves $v_t \ll K$
 122 when many actions share a state. The advantage over action-level weighting disappears only in the
 123 stronger no-sharing case of a one-to-one deterministic map with $K = |\mathcal{S}|$, where $v_t = K$. Against
 124 $\tilde{O}(\sqrt{(K + d)T})$, the rate available once the state-to-loss map is unrestricted (Esposito et al., 2023),
 125 the guarantee improves when $\bar{v}(d + 1) \log K \lesssim K + d$, that is when many actions are resolved by
 126 comparatively few shared states, and outside that regime it can be worse. Four limitations remain.

The delay is charged multiplicatively. The bound carries v_t where undelayed mediator feedback pays $v_t - 1$, so it does not vanish when the rows of P agree and the regret is zero. It carries no E_2 and so cannot approach the drift term of Theorem 3. And where the best action reaches the single best state a simple heuristic does better, at linear regret when it does not (Appendix I).

P need not be known, since (A_t, S_t) arrives undelayed, and the difficulty there looks like certification rather than estimation (Appendix E).

5 Choosing the representation

If the price of pooling is an effective dimension, the states become a design choice, finer ones raising it and coarser ones lowering it while blurring losses together. Fix $g : \mathcal{S} \rightarrow \mathcal{Z}$ and run the same algorithm on $g(S_t)$ with $P^g(z | a) = \sum_{s: g(s)=z} P(s | a)$. Coarsening never raises the dimension, $v_t(g) \leq v_t$, but the grouped estimator tracks the group average of θ_t and so acquires a bias. Write $\delta_t(g) = \max_z \max_{s, s': g(s)=g(s')=z} |\theta_t(s) - \theta_t(s')|$ for the widest within-group spread.

Theorem 2. *Under the hypotheses of Theorem 1, the algorithm run on g with $\sum_t v_t(g) \leq V(g)$ satisfies $\mathbb{E}[R_T] \leq \sqrt{2(d+1)V(g) \log K} + 2 \sum_t \delta_t(g)$.*

The terms move oppositely as groups coarsen, so the useful representations are the coarsest loss-homogeneous ones and the identity map recovers Theorem 1. The representation is fixed before the run and the theorem gives no rule for choosing one.

6 The cost of drift

The previous sections ask how the structural channel reduces the cost of estimation. We now ask how much drift costs even when the stale action losses are supplied for free.

Give each of $q \leq \min\{K-1, |\mathcal{S}|-1\}$ actions a private state, send the rest to one inert state, and drive the private states by independent Rademacher signs of amplitude ε held for blocks of h rounds (Appendix G). Private states stop pooling from helping, forcing $q \leq |\mathcal{S}|-1$.

If $h \leq d$ then A_t is independent of the signs governing its own block, even when the learner holds m_t exactly, since every usable outcome predates the block and a block never recurs (Lemma 14, Appendix G).

Theorem 3. *Fix T , $1 \leq d \leq T/2$, $E \leq T/4$ and $1 \leq q \leq \min\{K-1, |\mathcal{S}|-1\}$. There is an oblivious instance with $E_2 \leq E$ on which every algorithm, including one given the exact stale vector, satisfies $\mathbb{E}[R_T] = \Omega(\sqrt{dE} \min\{1 + \log q, T/d\})$. For a learner without that oracle and with $K \leq T$, the model class additionally forces $\mathbb{E}[R_T] = \Omega(\sqrt{KT})$, so the unhinted rate is $\Omega(\sqrt{KT} + \sqrt{dE} \min\{1 + \log q, T/d\})$.*

The drift term is the extra cost of acting on stale predictions, with d the rounds information is unavailable, E the accumulated squared error and q the coordinates the comparator can exploit. The $\sqrt{KT}$ term is the separate cost of finding a good action and is not oracle-robust, since m_t collapses the instance forcing it. The two are proved on separate instances, so the sum holds up to a constant, and the bound describes the family rather than separating algorithms.

7 Discussion

Two channels, two results, one gap. The structural channel has an algorithm, a rate in the effective dimension and a representation trade-off, all for known P . The temporal channel has a lower bound no algorithm avoids even given the exact stale vector. They do not meet because STATE-EXP3 never reads m_t and the lower bound says nothing about the rows of P . For each strengthening below the natural route is ruled out but not refuted (Appendices H, J).

Conjecture 4. *STATE-EXP3 at $m = 1$ satisfies the first bound, and some algorithm the second.*

$$\mathbb{E}[R_T] = \tilde{O}(\sqrt{V \log K} + \sqrt{dT \log K}), \quad \mathbb{E}[R_T] = \tilde{O}(\sqrt{KT} + \sqrt{dE_2 \log K}).$$

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A Extended related work

The minimax characterization of delayed adversarial bandits traces to Cesa-Bianchi et al. (2019), and Joulani et al. (2013) give the additive form of the delay penalty in the stochastic case. On acting before the loss is known, Flaspohler et al. (2021) study optimism under delay directly, Wei et al. (2020) delimit when prediction error controls regret, Hsieh et al. (2022) show that an optimistic step must run at a larger scale than the update, and Zhang (2026) gives second-order path-length bounds.

Structure that lets one observation inform several actions is the other half of the picture. Feedback graphs (Alon et al., 2015) fix which actions are revealed together, and their stochastic version (Esposito et al., 2022) randomizes that set. Factored rewards with intermediate observations (Mussi et al., 2024) decompose the reward instead. Off-policy coverage (Gabbianelli et al., 2023) is the closest analogue of our effective dimension, since its mismatch coefficient measures the same kind of overlap between the distribution that generated the data and the one being evaluated. The distinction from all of these is that a feedback graph randomizes *who* is observed, whereas our sensor randomizes *what* is observed and the sharing is a consequence of P rather than a modeling primitive.

B Notation

Table 1: Symbols used in the paper. Every symbol is also defined at first use in the body, so this table is a reference rather than a dependency.

Symbol	Meaning
T, K, d	rounds, actions, fixed feedback delay
$\mathcal{S}, S_t$	finite state space, state observed at round t
$P \in [0, 1]^{K \times \mathcal{S} }$	row-stochastic action-to-state matrix, fixed known in Theorems 1 and 2, estimated in Appendix E
$\theta_t \in [0, 1]^{ \mathcal{S} }$	state-loss means at round t , oblivious
$c_t = P\theta_t \in [0, 1]^K$	induced loss over actions
$q_t(s) = \sum_a x_t(a)P(s a)$	state distribution induced by the play
$X_t \in [0, 1]$	outcome at time $t + d$, $\mathbb{E}[X_t \mathcal{F}_{t-1}, A_t, S_t] = \theta_t(S_t)$
x_t, A_t	action distribution at round t , action drawn from it
$m_t = c_{t-d}$	stale action-loss vector, given before the action, $m_t = c_1$ for $t \leq d$
E_2	$\sum_t \ c_t - m_t\ _\infty^2$, accumulated squared prediction error
Λ_2	$\sum_t \min\{1, \ c_t - m_t\ _2^2\}$, the measure of Bar-On and Mansour (2025)
W	$\sum_t \ \theta_t - \theta_{t-1}\ _\infty^2$, state-loss variation
$\hat{c}_t, \tilde{c}_t, \check{c}_t$	state-pooled, action-level, centered loss estimates
$\hat{g}_t$	the Safe-Pool blend $(1 - \lambda_t)\hat{c}_t + \lambda_t\check{c}_t$, a distinct object
$\bar{c}_t, c_t^g, b_t$	conditional mean of $\hat{c}_t$, grouped surrogate loss, $\bar{c}_t - c_t$
$v_t, \bar{v}, V$	effective dimension $\sum_s q_t(s)^{-1} \sum_a x_t(a)P(s a)^2$, its supremum, its budget
$g, \mathcal{Z}, \delta_t(g)$	state grouping, its range, widest within-group loss spread
E, q	budget constraining E_2 , number of independently drifting coordinates
$h, B, \varepsilon, \sigma_b^{(j)}$	block length, block count, amplitude, Rademacher signs
η, β, C	learning rate, Tsallis parameter, residual scale
α	smoothing pseudocount in $\hat{P}_t$
κ	sensor balance, $\bar{p}(s) \geq 1/(\kappa \mathcal{S})$ for $\bar{p} = K^{-1} \sum_a P(\cdot a)$
m, L_i, G_r	interleaving factor, estimate of copy i , estimates outstanding at r
$D, d_{\max}, \bar{\Delta}, \Gamma$	total delay, maximal delay, inaccuracy bound, the measure of Wang et al. (2026)

C The state-pooled estimator

The algorithm. Inputs are the action-to-state matrix P , the delay d , the number of copies m , and the rate $\eta > 0$. The analyzed setting is $m = d + 1$ and the practical variant is $m = 1$. Round r 's outcome becomes usable at round $r + d + 1$, which for $m = d + 1$ is the next round belonging to the copy that played round r .

1. Set $L_i \leftarrow (0, \dots, 0) \in \mathbb{R}^K$ for $i = 0, \dots, m - 1$, and let $\mathcal{P}$ be an empty table of rounds awaiting their outcome.
2. For $t = 1, \dots, T$:
 - (a) *Deliver.* If $r \triangleq t - d - 1 \geq 1$, read $(i_r, q_r(S_r), S_r, X_r)$ from $\mathcal{P}$, remove it, and update only that copy, $L_{i_r}(a) \leftarrow L_{i_r}(a) + P(S_r | a)X_r/q_r(S_r)$ for every a . This is the pooled step, and it touches every action rather than the one played.
 - (b) *Select the copy.* $i \leftarrow t \bmod m$.
 - (c) *Play.* $x_t(a) \propto \exp(-\eta L_i(a))$, draw $A_t \sim x_t$, and observe S_t at once.
 - (d) *Record.* Compute the scalar $q_t(S_t) = \sum_a x_t(a)P(S_t | a)$ and store $(i, q_t(S_t), S_t, \cdot)$ in $\mathcal{P}$, to be completed when X_t arrives at time $t + d$.

Steps (a), (c) and (d) are each $O(K)$ and there is no optimization subproblem, so a round costs $O(K)$ once P is stored. The state is held in the m vectors L_i , which is mK numbers, the matrix P , which is $K|\mathcal{S}|$ numbers, and four scalars for each of the at most d rounds in flight. Only the scalar $q_r(S_r)$ is needed from round r , not the vector x_r .

Remark 5 (The χ^2 form). Under x_t the pair (A_t, S_t) has law $x_t(a)P(s | a)$ with marginals x_t and q_t , so

$$\begin{aligned}\chi^2(P_{A_t, S_t} \parallel P_{A_t} \otimes P_{S_t}) &= \sum_{a, s} \frac{(x_t(a)P(s | a) - x_t(a)q_t(s))^2}{x_t(a)q_t(s)} \\ &= \sum_{a, s} \frac{x_t(a)P(s | a)^2}{q_t(s)} - 2 + 1 = v_t - 1,\end{aligned}$$

the cross term summing to $\sum_{a, s} x_t(a)P(s | a) = 1$ and the last to $\sum_{a, s} x_t(a)q_t(s) = 1$. So $v_t = 1$ exactly when A_t and S_t are independent, which is when the rows of P agree, and the upper endpoint $|\mathcal{S}|$ is attained when P is deterministic and onto. This is an identity, not a bound, and it is used here only for reading v_t , never in a proof.

Proposition 6. Fix t , let x_t be the play distribution, $q_t(s) = \sum_a x_t(a)P(s | a)$ the induced state distribution, and $\hat{c}_t(a) = P(S_t | a)X_t/q_t(S_t)$, which is well defined because only states with $q_t(s) > 0$ occur. Then $\hat{c}_t(a) \geq 0$, $\hat{c}_t(a) \leq 1/x_t(a)$, $\mathbb{E}[\hat{c}_t(a) | \mathcal{F}_{t-1}] = c_t(a)$, and

$$\sum_a x_t(a)\mathbb{E}[\hat{c}_t(a)^2 | \mathcal{F}_{t-1}] \leq v_t := \sum_{s: q_t(s) > 0} \frac{\sum_a x_t(a)P(s | a)^2}{q_t(s)}, \quad 1 \leq v_t \leq |\mathcal{S}|,$$

states of zero probability being omitted throughout, which is the convention $0/0 = 0$. The action-level $\tilde{c}_t(a) = \mathbf{1}\{A_t = a\}X_t/x_t(a)$ has the same first three properties and weighted second moment at most K .

Proof. Nonnegativity is immediate from $X_t \geq 0$. Given $\mathcal{F}_{t-1}$ the state S_t has distribution q_t , and $\mathbb{E}[X_t | \mathcal{F}_{t-1}, S_t = s] = \theta_t(s)$ by the model, since that conditional mean does not depend on the action, so $\mathbb{E}[\hat{c}_t(a) | \mathcal{F}_{t-1}] = \sum_s q_t(s)P(s | a)\theta_t(s)/q_t(s) = c_t(a)$. For the range, $q_t(s) \geq x_t(a)P(s | a)$ for every a , so $P(s | a)/q_t(s) \leq 1/x_t(a)$ and $X_t \leq 1$. For the second moment, $X_t^2 \leq 1$ gives $\mathbb{E}[\hat{c}_t(a)^2 | \mathcal{F}_{t-1}] \leq \sum_s P(s | a)^2/q_t(s)$, and exchanging the sums gives v_t . Each summand of v_t is at most 1, since $P(s | a) \leq 1$ makes $\sum_a x_t(a)P(s | a)^2 \leq q_t(s)$, and at least $q_t(s)$, since Jensen applied to $a \mapsto P(s | a)$ under x_t gives $\sum_a x_t(a)P(s | a)^2 \geq q_t(s)^2$. Summing the two bounds gives $1 \leq v_t \leq |\mathcal{S}|$. The action-level claims are the same computation with the state replaced by the played action, whose weighted second moment is $\sum_a x_t(a)/x_t(a) = K$. $\square$

The two ends are attained. If every action induces the same state distribution then $v_t = \sum_s q_t(s) = 1$, and if P is deterministic, each action reaching a single state, then every $P(s | a)$ is 0 or 1 and each summand is exactly 1, so v_t counts the states reached and equals $|\mathcal{S}|$ when every state has an action reaching it. Disjoint row supports give $v_t = K$ rather than $|\mathcal{S}|$, since the summands over one action's private states add to $\sum_s P(s | a) = 1$, so a sensor with many private states per action resolves few of them. So v_t reads the overlap between the rows of P rather than their length, and $\bar{v} = \sup_x v(x)$, the supremum over play distributions, depends on P alone. An outcome observed at a state updates the estimate for every action that can reach it, which is why the construction of Section 6 gives each drifting coordinate a private state, forcing $v_t = |\mathcal{S}|$ there. The immediately observed pairs (A_t, S_t) identify P separately, though Theorem 1 assumes it known.

D Proofs of Theorems 1 and 2

Proof of Theorem 1. Fix a copy i and let T_i be the number of rounds it owns, so $\sum_i T_i = T$. Round t 's outcome is usable from round $t + d + 1$, which with $m = d + 1$ is exactly $t + m$, the copy's next round, so within a copy the feedback is undelayed and L_i carries exactly the estimates of that copy's earlier rounds. Exponential weights on nonnegative losses give, for any action a^* and any $\eta > 0$,

$$\sum_{t \in i} (\langle x_t, \hat{c}_t \rangle - \hat{c}_t(a^*)) \leq \frac{\log K}{\eta} + \frac{\eta}{2} \sum_{t \in i} \sum_a x_t(a) \hat{c}_t(a)^2,$$

by the usual potential telescoping with $e^{-z} \leq 1 - z + z^2/2$, which needs $z \geq 0$ and therefore $\hat{c}_t \geq 0$, supplied by Proposition 6. Each x_t is $\mathcal{F}_{t-1}$ -measurable, because L_i then holds only rounds $r \leq t - m = t - d - 1$, so taking expectations turns the left side into $\sum_{t \in i} (\langle x_t, c_t \rangle - c_t(a^*))$ by unbiasedness and the quadratic term into at most $\mathbb{E}[\sum_{t \in i} v_t]$, again by Proposition 6. Summing over the $m = d+1$ copies and using $\sum_i \min_a \sum_{t \in i} c_t(a) \leq \min_a \sum_t c_t(a)$, which is what lets the copies be compared against one common action, gives $\mathbb{E}[R_T] \leq (d+1) \log K/\eta + \frac{\eta}{2} \mathbb{E}[\sum_t v_t]$. If $\sum_t v_t \leq V$ surely then $\eta = \sqrt{2(d+1) \log K/V}$ balances the two terms and gives $\sqrt{2(d+1)V \log K}$, and $V = \bar{v}T$ is admissible by the definition of $\bar{v}$. $\square$

The delay enters as a factor because the copies do not share their estimates, each paying $\log K/\eta$ over $T/(d+1)$ rounds. Sharing them is exactly the $m = 1$ algorithm of Conjecture 4, whose analysis is open.

Lemma 7 (Coarsening). *For every play distribution and every g , $v_t(g) \leq v_t$.*

Proof. It suffices to merge two states, the general case following by induction on the merges, and a merged group of zero probability contributes nothing on either side by the convention above. Write P_1, P_2 for the two columns, $A_j = \sum_a x_t(a)P_j(a)^2$, $B_j = \sum_a x_t(a)P_j(a)$ and $\lambda = B_1/(B_1 + B_2)$. By Cauchy–Schwarz, $(P_1 + P_2)^2 \leq \lambda^{-1}P_1^2 + (1-\lambda)^{-1}P_2^2$ pointwise in a , so $\sum_a x_t(a)(P_1 + P_2)^2 \leq (B_1 + B_2)(A_1/B_1 + A_2/B_2)$, and dividing by the merged denominator $B_1 + B_2$ shows the merged summand is at most $A_1/B_1 + A_2/B_2$, the two it replaces. $\square$

Proof of Theorem 2. Running the algorithm on g is running it on the sensor $(\mathcal{Z}, P^g)$. The surrogate losses below depend on x_t and so are predictable rather than oblivious, but the proof of Theorem 1 uses obliviousness nowhere, only that each c_t^g is $\mathcal{F}_{t-1}$ -measurable given the play, so it applies unchanged and bounds regret measured in the surrogate losses $c_t^g(a) = \sum_z P^g(z | a)\bar{\theta}_t(z)$, where $\bar{\theta}_t(z) = \sum_{s: g(s)=z} q_t(s)\theta_t(s)/q_t^g(z)$ is the group mean the grouped estimator is unbiased for, by the same computation as in Proposition 6 with S_t replaced by $g(S_t)$. Fix a and z . Both $\{q_t(s)/q_t^g(z)\}$ and $\{P(s | a)/P^g(z | a)\}$ are probability vectors on $\{s : g(s) = z\}$, so their θ_t -averages lie in a common interval of length $\delta_t(g)$ and differ by at most $\delta_t(g)$. Weighting by $P^g(z | a)$ and summing over z gives $|c_t^g(a) - c_t(a)| \leq \delta_t(g)$ for every a , hence $\langle x_t, c_t \rangle - c_t(a^*) \leq \langle x_t, c_t^g \rangle - c_t^g(a^*) + 2\delta_t(g)$. Summing over t adds $2 \sum_t \delta_t(g)$, and Lemma 7 makes $V(g)$ no larger than a bound valid for the raw states. $\square$

E An unknown action-to-state map

The plug-in estimator. The pairs (A_t, S_t) arrive without delay, so P is estimable from the learner’s own play while the outcomes are still in flight. Let $N_a(t)$ count the rounds before t that played a and $N_{a,s}(t)$ those that also produced s , and set $\hat{P}_t(s | a) = (N_{a,s}(t) + \alpha)/(N_a(t) + \alpha|\mathcal{S}|)$ for a smoothing constant $\alpha \in (0, 1]$ fixed independently of T , which is row-stochastic with every entry positive, so $\hat{q}_t = x_t^\top \hat{P}_t$ is positive too, and both are $\mathcal{F}_{t-1}$ -measurable. Smoothing moves an entry from the empirical row by at most $\alpha(|\mathcal{S}| - 1)/(N_a(t) + \alpha|\mathcal{S}|)$, attained when every observation of a sits on one state, and moves the row in ℓ_1 by at most twice that. Both are of lower order than the sampling term at fixed α and must be carried otherwise, and Appendix I shows the choice matters in practice. The plug-in estimate is $\hat{c}_t(a) = \hat{P}_t(S_t | a)X_t/\hat{q}_t(S_t)$. Write $\varepsilon_t = \max_a \|\hat{P}_t(\cdot | a) - P(\cdot | a)\|_1$ for its accuracy and $\rho_t = \max_{s: q_t(s) > 0} |\hat{q}_t(s) - q_t(s)|/q_t(s)$ for the relative error it induces on the state distribution, states of zero probability under the play being omitted since the estimator never charges them.

Proposition 8. *On $\{\rho_t \leq \frac{1}{2}\}$ the plug-in estimate satisfies $\hat{c}_t \geq 0$ and $\sum_a x_t(a)\mathbb{E}[\hat{c}_t(a)^2 | \mathcal{F}_{t-1}] \leq 2\hat{v}_t \leq 2|\mathcal{S}|$, and its conditional mean $\bar{c}_t$ satisfies $\sum_a x_t(a)(\bar{c}_t(a) - c_t(a)) = 0$ exactly, for every $\hat{P}_t$ and with no hypothesis at all, and $|\bar{c}_t(a) - c_t(a)| \leq 2\varepsilon_t(a) + 2\rho_t$ for every a , where $\varepsilon_t(a) = \|\hat{P}_t(\cdot | a) - P(\cdot | a)\|_1$.*

Proof. Write $\Delta_t = \hat{P}_t - P$ and $\delta_t = \hat{q}_t - q_t = \sum_a x_t(a)\Delta_t(\cdot | a)$. Substituting these into $\bar{c}_t(a) = \sum_s q_t(s)\hat{P}_t(s | a)\theta_t(s)/\hat{q}_t(s)$ and cancelling gives $\bar{c}_t(a) - c_t(a) = \sum_s \theta_t(s)[q_t(s)\Delta_t(s |$

324 $a) - P(s | a)\delta_t(s)]/\hat{q}_t(s)$, so with $\theta_t \in [0, 1]^{|S|}$,

$$|\bar{c}_t(a) - c_t(a)| \leq \sum_s \frac{q_t(s)|\Delta_t(s | a)| + P(s | a)|\delta_t(s)|}{\hat{q}_t(s)}.$$

325 The play-weighted identity needs no hypothesis. By definition $\sum_a x_t(a)\hat{P}_t(s | a) = \hat{q}_t(s)$, so
 326 $\sum_a x_t(a)\hat{c}_t(a) = X_t$ whatever $\hat{P}_t$ is, and taking conditional expectations gives $\sum_a x_t(a)\bar{c}_t(a) =$
 327 $\sum_s q_t(s)\theta_t(s) = \langle x_t, c_t \rangle$. On $\{\rho_t \leq \frac{1}{2}\}$ we have $q_t \leq 2\hat{q}_t$, and for the per-action bias the
 328 first sum of the display is at most $2\|\Delta_t(\cdot | a)\|_1 = 2\varepsilon_t(a)$ and the second at most $2\sum_s P(s |$
 329 $a)|\delta_t(s)|/q_t(s) \leq 2\rho_t$, because $P(\cdot | a)$ is a probability vector. The play-weighted identity
 330 does *not* survive reweighting: at $x_t = (\frac{1}{2}, \frac{1}{2})$, $P = I$, $\hat{P} = (\frac{9}{2}, \frac{1}{8})$ and $\theta_t = (0, 1)$ the
 331 bias is $(\frac{1}{9}, -\frac{1}{9})$ with zero x_t -average, while weights $(1, 0)$ leave $\frac{1}{18}$. For the second moment,
 332 $\sum_a x_t(a)\mathbb{E}[\hat{c}_t(a)^2 | \mathcal{F}_{t-1}] \leq \sum_s \frac{q_t(s)}{\hat{q}_t(s)} \cdot \frac{\sum_a x_t(a)\hat{P}_t(s|a)^2}{\hat{q}_t(s)} \leq 2\hat{v}_t$, and $\hat{v}_t \leq |S|$ by Proposition 6
 333 applied to $\hat{P}_t$. $\square$

334 **The warm-start guarantee.** WARM-START STATE-EXP3 plays uniformly for n_0 rounds, forms
 335 $\hat{P} = \hat{P}_{n_0+1}$ from those rounds alone, discards the cumulative estimates together with every outcome
 336 of a warm-start round that has not yet arrived, and then runs STATE-EXP3 with $m = d + 1$, that
 337 frozen $\hat{P}$, and each play mixed with the uniform action at rate γ . Freezing makes $\hat{P}$ measurable with
 338 respect to the first phase, so ε and ρ are fixed before the second phase begins, and discarding is what
 339 keeps the first phase's inaccurate updates out of every later distribution.

340 **Theorem 9.** Suppose $\bar{p}(s) \geq 1/(\kappa|S|)$ for every s , fix $0 < \gamma \leq \frac{1}{2}$, and take $n_0 \geq$
 341 $c\gamma^{-2}K\kappa^2|S|^2 \log(K|S|T)$ for a universal c . Then WARM-START STATE-EXP3 satisfies, for ev-
 342 every $\eta > 0$,

$$\mathbb{E}[R_T] \leq n_0 + 1 + \gamma T + \frac{(d+1)\log K}{\eta} + 2\eta|S|T + T(4\bar{\varepsilon} + 4\bar{\rho}),$$

343 where $\bar{\varepsilon} = \tilde{O}(\sqrt{K|S|/n_0})$ and $\bar{\rho} = \tilde{O}(\kappa|S|\sqrt{K/n_0}/\gamma) \leq \frac{1}{2}$ bound ε and ρ on an event of proba-
 344 bility at least $1 - 1/T$. Tuning η and then $\gamma \asymp (\kappa|S|)^{1/2}(K/n_0)^{1/4}$ and $n_0 \asymp T^{4/5}(\kappa|S|)^{2/5}K^{1/5}$
 345 gives $\mathbb{E}[R_T] = \tilde{O}(\sqrt{d|S|T \log K} + (\kappa|S|)^{2/5}K^{1/5}T^{4/5})$, provided $T = \tilde{\Omega}(K(\kappa|S|)^2)$, which is
 346 what makes that n_0 satisfy both $n_0 \leq T - d$ and the lower bound above with some $\gamma \leq \frac{1}{2}$. Outside
 347 that regime the tuning is vacuous and $R_T \leq T$ is the statement.

348 *Proof.* The first phase has losses in $[0, 1]$, so it contributes at most n_0 . Let G be the event that
 349 $N_a(n_0 + 1) \geq n_0/(2K)$ for every a , that $\|\hat{P}(\cdot | a) - P(\cdot | a)\|_1 \leq \bar{\varepsilon}$ for every a , and that
 350 $\max_{s,a} |\hat{P}(s | a) - P(s | a)| \leq \bar{\rho}\gamma/(\kappa|S|)$. Uniform play gives $\mathbb{E}[N_a] = n_0/K$, so a multiplicative
 351 Chernoff bound, the ℓ_1 deviation bound for a multinomial, Hoeffding, and a union bound over the
 352 $K|S|$ entries put $\Pr(G^c) \leq 1/T$ at the stated n_0 , the pseudo-counts adding $2\alpha(|S| - 1)/(N_a + \alpha|S|)$
 353 to each ℓ_1 deviation and half that to each entry, of lower order at fixed α . Off G bound the second
 354 phase by T , contributing at most 1.

355 On G the quantity $\hat{P}$ is measurable with respect to the first phase, and mixing gives $q_t(s) \geq \gamma\bar{p}(s) \geq$
 356 $\gamma/(\kappa|S|)$ at every later round, so $\rho_t \leq \kappa|S|\max_{s,a} |\hat{P}(s | a) - P(s | a)|/\gamma \leq \bar{\rho} \leq \frac{1}{2}$ for every
 357 $t > n_0$ and every realization of the second phase. Proposition 8 therefore applies at every such
 358 round, and conditionally on the first phase the second is exactly the algorithm of Theorem 1 driven
 359 by a nonnegative estimate with weighted second moment at most $2\hat{v}_t \leq 2|S|$. Its proof gives
 360 $\mathbb{E}[\sum_{t>n_0} (\langle p_t, \bar{c}_t \rangle - \bar{c}_t(a^*))] \leq (d+1)\log K/\eta + 2\eta|S|T$, the extra factor of two coming from
 361 $p_t \leq x_t/(1-\gamma) \leq 2x_t$. Playing x_t rather than p_t costs at most γ per round since $c_t \in [0, 1]^K$, so
 362 it remains to pass from $\bar{c}_t$ to c_t under p_t . Proposition 8 gives $\langle x_t, b_t \rangle = 0$ for the per-action bias b_t ,
 363 and $x_t = (1-\gamma)p_t + \gamma u$ for the uniform u , so $\langle p_t, b_t \rangle = -\frac{\gamma}{1-\gamma}\langle u, b_t \rangle$, which for $\gamma \leq \frac{1}{2}$ is at most
 364 $\max_a |b_t(a)|$ in absolute value. Each round therefore adds $|\langle p_t, b_t \rangle| + |b_t(a^*)| \leq 2\max_a |b_t(a)| \leq$
 365 $4\bar{\varepsilon} + 4\bar{\rho}$. Collecting the four contributions gives the display. For the rate, balancing γT against $2T\bar{\rho}$
 366 gives the stated γ and a combined $\tilde{O}(T(\kappa|S|)^{1/2}(K/n_0)^{1/4})$, and balancing that against n_0 gives
 367 the stated n_0 , under which $T\bar{\varepsilon}$ and the constraint on n_0 are of lower order. $\square$

368 **Safe blending.** Blending the plug-in with the action-level estimate makes the guarantee safe. Let
 369 $\lambda_t \in [0, 1]^K$ be $\mathcal{F}_{t-1}$ -measurable and put $\check{g}_t(a) = (1 - \lambda_t(a))\bar{c}_t(a) + \lambda_t(a)\hat{c}_t(a)$.

370 **Proposition 10.** $\check{g}_t \geq 0$, its conditional mean satisfies $\mathbb{E}[\check{g}_t(a) \mid \mathcal{F}_{t-1}] - c_t(a) = \lambda_t(a)(\bar{c}_t(a) -$
 371 $c_t(a))$ for the plug-in's conditional mean $\bar{c}_t$ of Proposition 8, and

$$\sum_a x_t(a) \mathbb{E}[\check{g}_t(a)^2 \mid \mathcal{F}_{t-1}] \leq (K - \Pi_t) + \hat{V}_t,$$

372 for $\Pi_t = \sum_a \lambda_t(a)$ and $\hat{V}_t = \sum_a \lambda_t(a) x_t(a) \mathbb{E}[\hat{c}_t(a)^2 \mid \mathcal{F}_{t-1}]$, which satisfies $\hat{V}_t \leq 2\hat{v}_t$ on $\{\rho_t \leq$
 373 $\frac{1}{2}\}$ and not in general, since the step $q_t/\hat{q}_t \leq 2$ behind it is exactly that event. We write $V_t^{\text{ub}}(\tau)$
 374 below for a computable upper bound on $\hat{V}_t$, reserving $\hat{V}_t$ for the conditional expectation itself.

375 *Proof.* Both parts are nonnegative and $\lambda_t(a) \in [0, 1]$. The mean identity is linearity together
 376 with $\mathbb{E}[\bar{c}_t(a) \mid \mathcal{F}_{t-1}] = c_t(a)$. For the second moment, $u \mapsto u^2$ is convex, so $\check{g}_t(a)^2 \leq$
 377 $(1 - \lambda_t(a))\bar{c}_t(a)^2 + \lambda_t(a)\hat{c}_t(a)^2$, and $x_t(a)\mathbb{E}[\bar{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \leq 1$ by Proposition 6. $\square$

378 So a weight of $\lambda_t(a)$ on the plug-in buys a variance saving of $\lambda_t(a)$ and pays a bias of $\lambda_t(a)$ times the
 379 plug-in's, and both endpoints are exact. Let $r_t(a)$ and $\hat{\rho}_t$ be any sequences computable from the im-
 380 mediate pairs that are *simultaneously valid*, meaning $\varepsilon_t(a) \leq r_t(a)$ for every a and $\rho_t \leq \hat{\rho}_t$ at every
 381 $t \leq T$, on an event of probability at least $1 - 1/T$. Write $V_t^{\text{ub}}(\tau) = 2 \sum_{a: r_t(a) \leq \tau} x_t(a) \sum_s \hat{P}_t(s \mid$
 382 $a)^2 / \hat{q}_t(s)$, which the learner can evaluate and which dominates $\hat{V}_t$ on $\{\rho_t \leq \frac{1}{2}\}$. SAFE-POOL
 383 STATE-EXP3 sets $\lambda_t \equiv 0$ when $\hat{\rho}_t > \frac{1}{2}$, and otherwise picks the best member of the candidate
 384 set $\{\perp\} \cup \{r_t(a) : a \leq K\}$, where $\perp$ stands for $\lambda_t \equiv 0$ and a numerical candidate τ stands for
 385 $\lambda_t(a) = \mathbf{1}\{r_t(a) \leq \tau\}$, best meaning smallest

$$\Psi_t(\tau) = \frac{\eta}{2} [(K - \Pi_t(\tau)) + V_t^{\text{ub}}(\tau)] + (4\tau + 4\hat{\rho}_t) \mathbf{1}\{\Pi_t(\tau) > 0\}, \quad \Psi_t(\perp) = \frac{\eta}{2} K.$$

386 Write Ψ_t^* for that smallest value. The candidate $\perp$ is listed apart from the numerical thresholds
 387 because a valid radius may equal zero, in which case $\tau = 0$ pools the actions attaining it rather than
 388 abstaining, so $\perp$ rather than $\tau = 0$ is what keeps $\frac{\eta}{2} K$ available on every round.

389 **Theorem 11.** For every $\eta > 0$, SAFE-POOL STATE-EXP3 with $m = d + 1$ satisfies

$$\mathbb{E}[R_T] \leq 1 + \frac{(d + 1) \log K}{\eta} + \mathbb{E}\left[\sum_{t=1}^T \Psi_t^*\right],$$

390 so $\eta = \sqrt{2(d + 1) \log K / (KT)}$ gives $\mathbb{E}[R_T] \leq 1 + \sqrt{2(d + 1)KT \log K}$ on every instance, which
 391 is the guarantee of action-level exponential weights, with an improvement on each round whose
 392 Ψ_t^* is strictly below $\frac{\eta}{2} K$. The certificate is computed from the learner's own observations and so
 393 is random, which is why the display carries an expectation. The tuned bound does not, because
 394 $\Psi_t^* \leq \frac{\eta}{2} K$ holds on every path.

395 *Proof.* On the validity event, of probability at least $1 - 1/T$ and contributing at most 1 off it, pooling
 396 happens only when $\hat{\rho}_t \leq \frac{1}{2}$ and hence only when $\rho_t \leq \frac{1}{2}$, which is what makes V_t^{ub} dominate the
 397 conditional second moment. Proposition 10 then supplies a nonnegative estimate whose weighted
 398 second moment is at most $(K - \Pi_t) + \hat{V}_t$, so the argument of Appendix D applies to the sur-
 399rogate losses. Passing to c_t adds both the played bias $|\sum_a x_t(a) \lambda_t(a) b_t(a)|$ and the comparator
 400 bias $|\lambda_t(a^*) b_t(a^*)|$, for $b_t = \bar{c}_t - c_t$. The cancellation of Proposition 8 is lost under per-action
 401 weights, as the counterexample there shows, so each is bounded separately. On a pooled action
 402 $\varepsilon_t(a) \leq r_t(a) \leq \tau$ gives $|b_t(a)| \leq 2\tau + 2\hat{\rho}_t$, and since $\sum_a x_t(a) \lambda_t(a) \leq 1$ and $\lambda_t(a^*) \leq 1$ the two
 403 together are at most $4\tau + 4\hat{\rho}_t$, charged only when something is pooled. The second-moment bound
 404 and the two bias terms are all functions of the observations, so the per-round costs they assemble
 405 into are the random Ψ_t^* and the sum is taken in expectation. For the tuned bound, $\perp$ belongs to the
 406 candidate set and $\Psi_t(\perp) = \frac{\eta}{2} K$, so $\Psi_t^* \leq \frac{\eta}{2} K$ on every path whatever the radii are. $\square$

407 **Why certification stays conservative.** The guarantee is safe and the algorithm is inert. Pooling
 408 is certified when $4(\tau + \hat{\rho}_t) \leq \frac{\eta}{2} (\Pi_t - V_t^{\text{ub}})$. At the tuned η both sides fall as $T^{-1/2}$, the left through
 409 $\tau \asymp \sqrt{K \log(K|\mathcal{S}|T)/T}$ under uniform play and the right through $\eta \asymp \sqrt{\log K / (KT)}$. Their ratio

is $\Theta(\sqrt{\log(K|\mathcal{S}|T)/\log K})$, which exceeds one and does not vanish with the horizon, growing slowly in T through the logarithmic factors, so no horizon makes the certificate clear. Appendix I reports behavior consistent with this explanation, the algorithm reproducing action-level regret to the digit with a pooled fraction of zero.

The obstruction is not the shape of the bias bound. Regret charges $\langle x_t, b_t \rangle - b_t(a^*)$ for the per-action bias b_t , while Proposition 8 charges $\max_a |b_t(a)|$. Measured against the budget itself at $K = 40$, $|\mathcal{S}| = 8$, $T = 8000$ and $d = 50$, the first is 0.42 times it and the second 1.67 times it, so on that trajectory the realized bias lies below the best-case variance budget and a tighter bias bound would close the analytical gap. Two exact refinements fail to deliver one. Both $\sum_s \Delta_t(s | a)$ and $\sum_s \delta_t(s)$ vanish, so a constant may be subtracted and a maximum replaced by an oscillation, and Cauchy-Schwarz against q_t replaces the comparator term by $\sqrt{\nu_t(a^*)} \chi_t$ for its own effective dimension ν_t and a chi-square χ_t between $\hat{q}_t$ and q_t . Measured, these are 6.1 and 63 times the truth against 3.9 for the maximum they replace.

What the algorithm can verify from its own counts is 104 times the budget, and Bernstein radii do not improve on Hoeffding ones (Appendix I), so the barrier is certifying ρ_t , the relative error the estimate induces on the state distribution. The play-weighted identity of Proposition 8 says the whole charge is the comparator's, and it routes through ρ_t only because $\hat{q}_t = x_t^\top \hat{P}_t$ predicts the state distribution out of every row of P . Predicting is expensive for that reason, needing $t = \Omega(K|\mathcal{S}| \log(K|\mathcal{S}|T)/\rho^2)$ rounds, and the resulting certificate clears for no $K \leq T$, since it would need $\log K \geq 16c|\mathcal{S}|$ for $c = \log(2|\mathcal{S}|T)$, which $\log K \leq \log T$ cannot meet.

Counting the states costs less and still buys nothing. With x_t frozen over a window the empirical state frequency is a plain multinomial estimate of q_t , needing $n = \Omega(|\mathcal{S}| \log(|\mathcal{S}|T)/\rho^2)$ samples, a factor K fewer than predicting and free of P , and the certificate does clear once $K \log K \geq 64c|\mathcal{S}|$, about $K \geq 10^3$ at $|\mathcal{S}| = 8$. The window that buys it must satisfy $n \geq 32c|\mathcal{S}|T/(K \log K)$, a constant fraction of the horizon, and freezing the play over blocks of length B costs $\sqrt{2TBK \log K}$, which at that B is $T\sqrt{128c|\mathcal{S}|}$ and so linear in T . Both routes fail for one reason: the certificate must resolve q_t to relative accuracy $\Theta(\sqrt{K \log K/T})$, finer than either estimate delivers from the data the learner has.

In words, an unknown action-to-state matrix costs a bias in two accuracies, the ℓ_1 error of $\hat{P}$ and the relative error it induces on the state distribution, and only the second needs forced exploration, which is what sets the rate. **The bound does not justify the channel it analyzes.** Its second term exceeds $\sqrt{KT}$ at every horizon with $T > K$, by a factor $(T/K)^{3/10}$, so a learner holding this guarantee alone should run action-level exponential weights instead. The experiments suggest the analysis is conservative rather than the algorithm weak, since Appendix I shows it tracking known P closely, and sharpening it is the natural next result.

F Feedback models, and what transfers

Write $\Lambda_2 = \sum_t \min\{1, \|c_t - m_t\|_2^2\}$ for the inaccuracy measure of Bar-On and Mansour (2025). Unlike E_2 , which charges only the worst action in a round, Λ_2 grows with how many actions are predicted wrongly, and since $\|v\|_\infty \leq \|v\|_2 \leq \sqrt{q}\|v\|_\infty$ for a vector v with q nonzero coordinates, $E_2 \leq \Lambda_2 \leq q E_2$ with both ends attained.

Bar-On and Mansour (2025) analyze a model in which the played coordinate of the loss vector is observed exactly. Section 3 instead observes a random outcome whose conditional mean is the loss. The following records precisely what survives the substitution.

Proposition 12. *Take the action-level $\tilde{c}_t(a) = \mathbf{1}\{A_t = a\}X_t/x_t(a)$ of Proposition 6 under the model of Section 3. Then for every action a , (i) $\mathbb{E}[\tilde{c}_t(a) | \mathcal{F}_{t-1}] = c_t(a)$, (ii) $0 \leq \tilde{c}_t(a) \leq 1/x_t(a)$, and (iii) $\mathbb{E}[\tilde{c}_t(a)^2 | \mathcal{F}_{t-1}] \leq 1/x_t(a)$. The centered estimator $\check{c}_t(a) = m_t(a) + \mathbf{1}\{A_t = a\}(X_t - m_t(a))/x_t(a)$ also satisfies (i), and its residual obeys $\mathbb{E}[(\check{c}_t(a) - m_t(a))^2 | \mathcal{F}_{t-1}] \leq 1/x_t(a)$.*

Proof. For (i), by the tower rule $\mathbb{E}[\mathbf{1}\{A_t = a\}X_t | \mathcal{F}_{t-1}] = x_t(a) \mathbb{E}[X_t | A_t = a] = x_t(a)c_t(a)$, and dividing by $x_t(a)$ gives the claim. Item (ii) holds because $X_t \in [0, 1]$. For (iii), $\mathbb{E}[\tilde{c}_t(a)^2 | \mathcal{F}_{t-1}] = \mathbb{E}[X_t^2 \mathbf{1}\{A_t = a\}]/x_t(a)^2 = \mathbb{E}[X_t^2 | A_t = a]/x_t(a) \leq 1/x_t(a)$ because $X_t \leq 1$. The statements for $\check{c}_t$ follow the same computation with X_t replaced by $X_t - m_t(a)$, using $|X_t - m_t(a)| \leq 1$. $\square$

462 In words, replacing an exactly observed loss by a bounded unbiased draw of it changes neither the
 463 mean of the estimator nor the moment bounds an analysis consumes.

464 Under exact feedback the corresponding estimator satisfies (i) to (iii) with identical bounds, since
 465 $c_t(a) \leq 1$. Consequently any step of an exact-feedback analysis that uses the observation only
 466 through unbiasedness and these two moment bounds is unchanged under noisy outcomes. What this
 467 does not cover is the comparison between the estimate built from the stale action-loss vector and the
 468 estimate built from the true loss. Under exact feedback that difference is determined by the played
 469 action. Under noisy outcomes it carries an additional mean-zero term of magnitude up to $1/x_t(A_t)$,
 470 and bounding its cumulative effect over a window of d rounds is what requires the stability property
 471 falsified in Appendix H. An upper bound for the noisy model is Theorem 1. What remains open is
 472 one adaptive to E_2 .

473 G Proofs

474 **Lemma 13** (Delay window). $E_2 \leq d^2 W$ for $W = \sum_t \|\theta_t - \theta_{t-1}\|_\infty^2$. The ratio $E_2/(d^2 W)$ tends
 475 to one when θ drifts monotonically in one coordinate at a constant rate and $T/d \rightarrow \infty$, the first d
 476 rounds contributing $\sum_{j < d} j^2$ rather than d^2 each under the convention $m_t = c_1$.

477 In words, accumulated squared prediction error is at most the state-loss variation inflated by the
 478 square of the delay, and no better bound in W alone is available.

479 *Proof of Lemma 13.* Throughout write $\Delta_j = \theta_j - \theta_{j-1}$, set $\theta_0 = \theta_1$, so $\Delta_1 = 0$, and read c_r as c_1 for
 480 $r \leq 0$, matching the convention $m_t = c_1$ for $t \leq d$ of Section 3. Then $c_t - c_{t-d} = P \sum_{j=t-d+1}^t \Delta_j$.
 481 Because the rows of P are probability vectors, averaging cannot increase the supremum norm, so
 482 $\|c_t - c_{t-d}\|_\infty \leq \sum_j \|\Delta_j\|_\infty$. By Cauchy–Schwarz applied to the d summands, $\|c_t - c_{t-d}\|_\infty^2 \leq$
 483 $d \sum_{j=t-d+1}^t \|\Delta_j\|_\infty^2$. Summing over t and using that each increment belongs to at most d windows
 484 gives $E_2 \leq d^2 W$. The ratio $E_2/(d^2 W)$ approaches one when every increment has the same mag-
 485 nitude, sign and active coordinate, and some action reaches that coordinate deterministically, since
 486 then no cancellation occurs within a full window and P passes the increment through undiminished.
 487 Without the second condition the drift can lie in the kernel of P and the ratio can be zero. $\square$

488 **Lemma 14** (Information isolation). If $h \leq d$ then A_t is independent of the signs governing its own
 489 block, and this remains true when the learner is provided with the exact stale action-loss vector m_t .

490 *Proof of Lemma 14.* For t in block b , every outcome available at t comes from a round $r \leq t - d \leq$
 491 $bh - d \leq (b-1)h$, by the definition of the delay and because $h \leq d$. Such rounds lie in blocks strictly
 492 before b , whose joint law depends only on $\sigma_2, \dots, \sigma_{b-1}$. Hence by conditional independence A_t is
 493 independent of block b 's signs. For the oracle, take any $h \leq d$. If $t > d$ then $t-d \leq bh-d \leq (b-1)h$
 494 by the same computation, so $m_t = c_{t-d}$ is measurable with respect to the signs of blocks strictly
 495 before b . If $t \leq d$ then $m_t = c_1 = \frac{1}{2}\mathbf{1}$, which is deterministic because block 1 is neutral. In both
 496 cases m_t is independent of block b 's signs, so adjoining it to the learner's information leaves the
 497 conclusion intact. $\square$

498 The construction in full. Take states $s_0, \dots, s_q$, a stationary deterministic map sending a_j to s_j for
 499 $j \leq q$ and every other action to s_0 , and independent Rademacher signs $\sigma_b^{(j)}$ for $2 \leq b \leq B = \lfloor T/h \rfloor$.
 500 Set $\theta_t(s_0) = \frac{1}{2}$ throughout, $\theta_t \equiv \frac{1}{2}$ on block 1, and $\theta_t(s_j) = \frac{1}{2} - \varepsilon \sigma_b^{(j)}$ on each later block b . The
 501 neutral first block makes $c_1 = \frac{1}{2}\mathbf{1}$, so the convention $m_t = c_1$ for $t \leq d$ of Section 3 supplies a
 502 constant over exactly the rounds that precede any usable feedback, which is what Lemma 14 needs
 503 at $b = 1$. Rounds after the last complete block carry no drift, costing a constant factor.

504 **Lemma 15** (Rademacher lower tail). Let S_B be a sum of $B \geq 32$ independent Rademacher signs.
 505 For every $1 \leq u \leq \sqrt{B}/32$, $\Pr(S_B \geq u\sqrt{B}) \geq \frac{u}{4} e^{-64u^2}$.

506 *Proof.* Write $S_B = 2X - B$ with $X \sim \text{Bin}(B, \frac{1}{2})$, so that $\{S_B \geq u\sqrt{B}\} = \{X \geq B/2 + u\sqrt{B}/2\}$.
 507 Passing to X removes the lattice gap, since X takes every integer value in $[0, B]$ whereas S_B takes

only those of the parity of B . Put $n = \lfloor B/2 \rfloor$ and $t = \lceil u\sqrt{B}/2 \rceil + 1$, so $X \geq n + t$ implies the event, and $t \leq u\sqrt{B}$ because $u\sqrt{B} \geq \sqrt{32} > 4$.

For $j \geq 1$ the ratio of consecutive binomial probabilities is $\Pr(X = n + j)/\Pr(X = n + j - 1) = (B - n - j + 1)/(n + j)$. Using $(B - 1)/2 \leq n \leq B/2$, its distance from one is at most $(2j - 1)/(n + j) \leq (4j - 2)/(B - 1) \leq 8j/B$ for $B \geq 2$. Restricting to $j \leq B/16$ keeps $8j/B \leq \frac{1}{2}$, where $\log(1 - x) \geq -2x$ holds, so summing over $i \leq j$ gives $\Pr(X = n + j) \geq \Pr(X = n)e^{-16j(j+1)/(2B)} \geq \Pr(X = n)e^{-16j^2/B}$. The central term satisfies $\Pr(X = n) \geq 1/(2\sqrt{B})$ for every $B \geq 2$.

Sum over the t integers $j \in \{t, \dots, 2t - 1\}$, all of which are at most $2t$ and hence at most $B/16$ because $u \leq \sqrt{B}/32$ forces $2t \leq 2u\sqrt{B} \leq B/16$. Each term is at least $e^{-64t^2/B}/(2\sqrt{B})$, and $t \leq u\sqrt{B}$ makes the exponent at most $64u^2$, so $\Pr(X \geq n + t) \geq te^{-64u^2}/(2\sqrt{B}) \geq \frac{u}{4}e^{-64u^2}$ using $t \geq u\sqrt{B}/2$. $\square$

Three regimes give the maximal inequality the lower bound needs. Each $S_B^{(j)}$ is sub-Gaussian with variance proxy B , so $\mathbb{E} \max_j (S_B^{(j)})^+ \leq \sqrt{2B \log q} + \sqrt{B}$ throughout. For the matching lower bound, first suppose $\log q \leq B/8$ and $\log q \geq 128$. Taking $u = \sqrt{\log q}/128$, which then lies in $[1, \sqrt{B}/32]$, makes $e^{-64u^2} = q^{-1/2}$, so $\Pr(S_B^{(j)} \geq u\sqrt{B}) \geq \frac{u}{4}q^{-1/2}$ and independence across j gives $\Pr(\max_j S_B^{(j)} < u\sqrt{B}) \leq e^{-u\sqrt{q}/4}$, at most $\frac{1}{2}$ beyond an absolute q . Hence $\mathbb{E} \max_j (S_B^{(j)})^+ \geq \frac{1}{2}u\sqrt{B} = \Omega(\sqrt{B \log q})$. Second, if $\log q > B/8$ and $B \geq 1024$, take $u = \sqrt{B}/32$, so $u\sqrt{B} = B/32$ and $e^{-64u^2} = e^{-B/16}$, and $q \geq e^{B/8}$ makes $qe^{-B/16}$ diverge, giving $\mathbb{E} \max_j (S_B^{(j)})^+ = \Omega(B)$, which is $\Omega(\sqrt{B \min\{1 + \log q, B\}})$ in that regime. Third, when $\log q < 128$ or $B < 1024$ the quantity $\sqrt{B \min\{1 + \log q, B\}}$ is $O(\sqrt{B})$, and a single walk already gives $\mathbb{E}(S_B^{(1)})^+ = \frac{1}{2}\mathbb{E}|S_B| \geq \frac{1}{2}(\mathbb{E}S_B^2)^{3/2}(\mathbb{E}S_B^4)^{-1/2} \geq \frac{1}{2}\sqrt{B/3}$ by Hölder, since $\mathbb{E}S_B^2 = B$ and $\mathbb{E}S_B^4 \leq 3B^2$. Combining the three regimes, $\mathbb{E} \max_j (S_B^{(j)})^+ = \Theta(\sqrt{B \min\{1 + \log q, B\}})$. This is the route by which maxima of independent random walks fix the minimax rate for prediction with expert advice (Cesa-Bianchi and Lugosi, 2006).

Proof of Theorem 3. Take $h = d$ and $\varepsilon = \frac{1}{2}\sqrt{E/T}$. By Lemma 14 every algorithm's expected loss equals $T/2$, since its action is independent of the current block's signs and all states have mean $\frac{1}{2}$ marginally. Its regret is therefore exactly the comparator's fluctuation advantage, $\varepsilon d \mathbb{E}[\max_{j \leq q} (\sum_{2 \leq b \leq B} \sigma_b^{(j)})^+]^+$ with $B = \lfloor T/d \rfloor$, over the $B - 1$ randomized blocks. Per round $\|c_t - m_t\|_\infty \in \{0, 2\varepsilon\}$, because the supremum norm charges only the largest coordinate and each coordinate moves by at most 2ε at a block boundary, so $E_2 \leq 4\varepsilon^2 T = E$ holds deterministically. The budget must bind on the realized sequence, since a budget holding only in expectation would not constrain the instance selected by Yao's principle. By Lemma 15 and the discussion following it, the expected maximum of the positive parts of q independent Rademacher sums of length $B - 1$ is $\Theta(\sqrt{(B - 1) \min\{1 + \log q, B - 1\}})$, which covers both regimes and the case $q = 1$. The hypothesis $d \leq T/2$ gives $B \geq 2$ and hence $B - 1 \geq B/2$, so replacing $B - 1$ by B costs a factor at most $\sqrt{2}$ in each branch of the minimum, including the saturated branch where the minimum is $B - 1$ rather than $1 + \log q$. Substituting $B = \lfloor T/d \rfloor$ and ε therefore gives $\Theta(\sqrt{dE \min\{1 + \log q, T/d\}})$. Rounds after the last complete block carry no drift and change the constant only. When the oracle is withheld, the $\sqrt{KT}$ term is Esposito et al. (2023, Thm. 5.1), whose construction is stationary and therefore lies in the present model class. $\square$

Proposition 16. Suppose $X_t = c_t(A_t)$. On instances whose drift is carried by a single action, $\Lambda_2 = E_2$, so the bound of Bar-On and Mansour (2025) reads $\tilde{O}(\sqrt{KT} + dE_2)$ and its drift contribution $\sqrt{dE_2}$ matches Theorem 3 up to logarithmic factors.

In words, single-action drift makes the Euclidean and supremum measures agree, so on that family the drift term of the published upper bound cannot be improved. The $\sqrt{KT}$ term is not matched, since a perfect hint leaves the loss vector known after the action and the problem full-information,

where the rate is $\Theta(\sqrt{T \log K})$. Neither statement covers noisy outcomes, for which no upper bound is available.

Proof of Proposition 16. If only one coordinate of $c_t - m_t$ is nonzero then $\|c_t - m_t\|_2 = \|c_t - m_t\|_\infty$. On the family of Section 6 that coordinate has magnitude at most $2\varepsilon \leq 1$, so the truncation at one in Λ_2 is inactive and $\Lambda_2 = E_2$. The bound of Bar-On and Mansour (2025) then reads $\tilde{O}(\sqrt{KT} + dE_2)$, whose $\sqrt{dE_2}$ contribution Theorem 3 matches up to the logarithmic factor at $q = 1$. Its $\sqrt{KT}$ contribution is not matched. $\square$

H The two-action counterexample

Take $K = 2$ with the hybrid regularizer of Zimmert and Seldin (2020), Tsallis parameter $\beta = 0.01$, learning rate $\eta = 0.5$ and residual scale $C = 1$. The iterate solves the stationarity condition $g(x_1) + L_1 = g(x_2) + L_2$ with $x_1 + x_2 = 1$, where

$$g(x) = -\frac{1}{2\beta}x^{-1/2} + \frac{1}{\eta} \log x$$

and L_a is the cumulative estimate for action a . Suppose the cumulative estimates place action 1 at probability x_1 . A single maximally negative centered residual $-C/x_1$ is then delivered to action 1. This is realizable in the model of Section 3, since $m_t(1) = 1$ and $X_t = 0$ give exactly $\tilde{c}_t(1) - m_t(1) = -C/x_1$ with $C = 1$. Solving the stationarity condition again gives the following.

x_1 before	x_1 after	ratio
9.15×10^{-3}	1.46×10^{-2}	1.6
9.83×10^{-4}	7.39×10^{-3}	7.5
2.19×10^{-4}	9.98×10^{-1}	4561
5.80×10^{-5}	1.00	17226
4.86×10^{-6}	1.00	205641

The transition sits at $4\beta^2 C^2 = 4 \times 10^{-4}$, which is where the Tsallis term $x^{-1/2}$ ceases to dominate the entropy term $\log x$ in g , so the one-step linearization underlying a constant-factor bound fails. The violated statement is the local-stability inequality asserting that $x_{t'}(a)/x_t(a)$ stays bounded by a constant for all t' within d rounds of t . A single round refutes it, so no accumulation argument is needed. This invalidates that lemma and the proof route resting on it, and not every argument that might use the same regularizer. It does not refute Conjecture 4, which concerns the regret rate and may be provable by other means. Devices that bound the residual magnitude introduce either a first-order bias dominating the target term or a probability floor narrowing the delay regime.

I Numerical detail

Fifteen studies, alongside Figure 2, which sweeps row overlap against delay in the three settings the studies treat separately, P known, P estimated, and states coarsened. Studies one to thirteen draw P from a Dirichlet and studies fourteen and fifteen do not. Study fifteen is the one where STATE-EXP3 is beaten. Each study is reported as an observation, an interpretation, and a limitation. Studies 8 and 9 ask whether the two theorems describe the right *dependence* rather than whether a bound holds, which is the question a loose bound leaves open. Study 10 is a control for study 6 and reverses its conclusion, and study 13 reports a rule that does not work. All policies run on the same environment parameters and common random-number streams, so a seed fixes everything except which estimator is formed, and differences are taken within a seed before averaging. Round r 's outcome is consumed before round $r + d + 1$, matching Section 3. Reported $\pm$ is one standard error of the paired difference.

Does the bound hold, and does the estimator help? With $|\mathcal{S}| = 4$, $T = 10000$, $d = 50$, drift band 0.05 and 100 seeds:

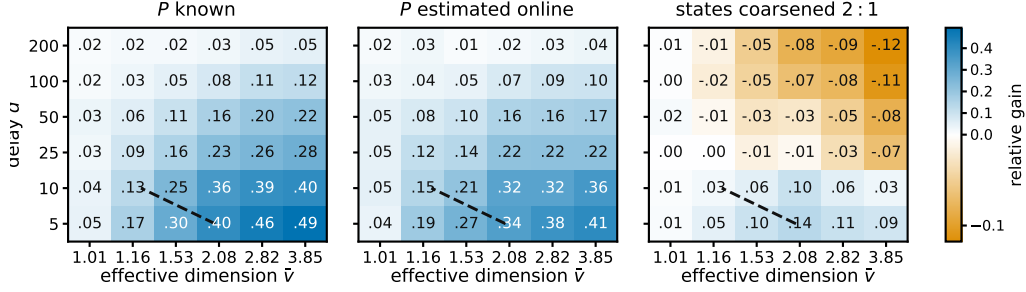


Figure 2: Relative gain $(R_{\text{action}} - R_{\text{state}})/R_{\text{action}}$ of pooling through the state, for the practical $m = 1$ variant, at $K = 40$, $|S| = 8$, $T = 5000$ and 12 paired seeds. Blue is a gain, and $\bar{v}$ is a Monte Carlo estimate over 4000 play distributions of the *raw* map, which for the coarsened panel is not the $\bar{v}(P^g)$ that Theorem 2 charges. The dashed line marks where the bound of Theorem 1, which analyzes $m = d + 1$, falls below $\tilde{O}(\sqrt{(K + d)T})$. It compares two upper bounds for a variant other than the one plotted, so it is a reference and not a prediction.

K	state, $m = d + 1$	state, $m = 1$	action, $m = 1$	paired gain	state better in
8	455.1	368.2	387.6	19.3 ± 3.4	73/100
20	618.3	493.7	547.5	53.8 ± 5.6	87/100
40	734.3	574.5	696.1	121.5 ± 9.4	99/100
80	780.4	614.8	784.0	169.3 ± 9.9	100/100

Observed. Realized regret for the analyzed variant stayed below $\sqrt{2(d+1)|S|T \log K}$ in 100 of 100 seeds at every K , the bound running from 2913 to 4228 against realized 455 to 780. The plain variant beat the action-level baseline at every K , by a margin growing from 19 to 169 as K rose tenfold with $|S|$ fixed, and it beat the analyzed variant throughout. Sweeping the delay at $K = 40$ gave state-pooled 387, 569 and 731 against action-level 653, 699 and 773 at $d = 10, 50$ and 200.

Interpretation. The margin growing in K at fixed $|S|$ is the prediction that separates the two estimators, so the state channel is doing the work the effective dimension attributes to it. Interleaving looks like a price of the analysis rather than of the estimator. The narrowing of the gap as the delay grows is what an additive delay term predicts and a multiplicative one does not, which is the diagnostic evidence behind Conjecture 4.

Limitation. This is one synthetic family with a fixed drift band, the plain variant carries no guarantee, and a simulation cannot establish that a bound holds, only that realized regret stayed below it in the runs performed.

What does an unknown map cost? What follows evaluates a practical variant, the plug-in estimate with $\hat{P}_t$ updated every round, no restart, $m = 1$ and $\alpha = 0.01$, and not the warm-start algorithm of Theorem 9, which freezes $\hat{P}$ and restarts. Against the same algorithm given P and against the action-level baseline, over 60 paired seeds and at mixing rate γ :

K	γ	known P	plug-in	action-level	plug-in gain
8	0.00	366.2	365.3	385.4	20.0 ± 4.3
20	0.00	480.3	477.9	529.0	51.2 ± 7.6
40	0.00	582.8	585.3	710.2	124.8 ± 11.2
80	0.00	625.2	623.1	802.2	179.1 ± 12.9
40	0.02	590.5	592.8	712.9	120.1 ± 10.4
40	0.10	618.9	623.5	735.5	112.0 ± 10.1

Observed. The plug-in landed within 0.5 per cent of known P at every K and beat action-level weighting by 20 to 179 as K rose tenfold. Forced exploration did not help, $\gamma = 0$ being best at every K . At $K = 40$ the sampled maps have $\kappa \approx 1.14$, so the exploration term $(\kappa|S|)^{2/5}K^{1/5}T^{4/5}$ of

615 Theorem 9 evaluates to about 6100 and its leading term to about 2700, against realized regret 585
616 and against $\sqrt{KT} = 632$.

617 Interpretation. Estimating P is nearly free on this family, and the gap between the analysis and the
618 behavior is an order of magnitude, so the analysis rather than the method looks conservative.

619 Limitation. The variant run here has no guarantee of its own, so the comparison is against a rate
620 proved for a different algorithm. That $\gamma = 0$ wins is a statement about this family rather than about
621 the algorithm, which the next study tests.

622 **Where does the plug-in fail?** The family above is mild. Five deliberately hostile maps at $K = 40$,
623 $|\mathcal{S}| = 8$, $T = 8000$, $d = 50$ and 30 paired seeds, with κ measured on each: a best action reachable
624 only through a state the others rarely produce, a sensor whose mass sits on one state, a state of
625 probability 10^{-3} everywhere, a deterministic map, and one whose good action's early visits are
626 unrepresentative.

environment	κ	known P	action-level	plug-in at mixing rate γ			
				0	0.01	0.05	0.20
rare optimal action	125.9	950.7	1302.3	948.2	976.8	1087.8	1503.0
unbalanced coverage	152.7	483.4	604.4	474.1	476.2	487.0	531.8
near-unreachable state	108.8	552.9	650.4	559.5	563.3	575.8	624.7
deterministic rows	2.5	556.3	733.7	571.9	585.9	653.9	904.3
misleading early visits	1.5	674.0	847.2	679.9	688.0	716.8	834.3

628 Observed. The plug-in beat action-level weighting on all five, including two with κ above 100, and
629 forced exploration cost regret in every cell. At the Laplace default $\alpha = 1$ the deterministic map is
630 the one case where the plug-in *loses* to action-level weighting, 750.3 against 733.7. Sweeping α
631 over 1, 0.3, 0.1, 0.03, 0.01 gives 750.3, 642.5, 601.4, 581.1, 571.9 there and is monotone on all five,
632 and $\alpha = 0.01$ is what the tables above use.

633 Interpretation. The mixing rate that Theorem 9 tunes is doing analytic work rather than algorithmic
634 work. The single loss has an identifiable mechanism, since one visit to a deterministic row is
635 recorded as $\hat{P}(s | a) = 2/(1 + |\mathcal{S}|)$ rather than 1, so rarely played actions are scored as if spread
636 over states they never reach.

637 Limitation. The leading rate is indifferent to fixed α and the algorithm is not, which is worth stat-
638 ing rather than tuning away. These five maps show that hostility to the plug-in is survivable at a
639 small constant, not that it is harmless, and no map adversarial to the smoothing constant itself was
640 constructed.

641 **Does safe blending recover the pooling gain, and if not why not?** The rule of Theorem 11, run
642 in the $m = 1$ variant like everything else in this appendix rather than in the analyzed $m = d + 1$
643 one, was compared against its two ingredients at $K = 40$, $|\mathcal{S}| = 8$, $d = 50$ and 20 paired seeds.
644 Measuring the realized bias against the certificate's own budget locates the outcome. At $K = 40$,
645 $|\mathcal{S}| = 8$, $T = 8000$, $d = 50$ and 12 seeds the per-round budget $\frac{\eta}{2}(K - 2|\mathcal{S}|)$ sums to 307.3 over the
646 horizon, and

summed	versus budget	quantity
128.7	0.42×	what the regret charges, $\sum_t \langle x_t, b_t \rangle - b_t(a^*) $
513.7	1.67×	what Proposition 8 charges, $\sum_t \max_a b_t(a) $
32000.0	104×	what the algorithm can verify, Hoeffding radii
32000.0	104×	the same with Bernstein radii

648 Observed. The rule reproduced action-level regret to the digit in every cell, 533.2, 754.4 and 1317.0
649 at $T = 8000$ on the Dirichlet, deterministic and rare-optimal families and 2304.6, 2023.5 and 3108.8
650 at $T = 50000$, against a plug-in that scores 464.8, 585.5, 956.5 and 1770.6, 1473.5, 2346.6 on the
651 same runs. The fraction of round-action pairs it cleared for pooling was 0.000 everywhere, at both
652 horizons, at both smoothing constants, and with the confidence logs dropped as well.

653 Interpretation. The implemented certificate selected no pooling in any run, so the policy coincided
654 numerically with the action-level baseline rather than merely matching its guarantee, and the table
655 says where that comes from. On this trajectory the realized bias lies below the best-case variance

budget, so pooling would pay for itself here. The bound of Proposition 8 is 1.67 times the budget with a factor of 4.0 available between it and the truth, so a tight bias bound would close the analytical gap. What the algorithm can verify is saturated at the trivial cap of four per round, and Bernstein radii buy nothing over Hoeffding ones, which is where the failure lives.

Limitation. Two exact refinements of the bias bound, an oscillation in place of a maximum and a Cauchy–Schwarz split into the comparator’s effective dimension times a chi-square, give 577.9 and 5931.6 against a truth of 94.1 over $T = 4000$, where the maximum gives 368.0, so both are worse than what they replace. The budget table is one diagnostic trajectory rather than a worst case, so it shows this certificate inert here and not that no certificate can clear.

Can the denominator be counted instead of predicted? With x frozen over blocks of 400 and the states of the first half counted, at $K = 40$, $|\mathcal{S}| = 8$, $T = 8000$, $d = 50$ and 12 seeds, the counted denominator gives regret 496.5 against 499.5 for the predicted one and a comparator bias of 99.6 against 52.9, both under the budget of 153.7 for the pooled half.

Observed. Counting is the noisier of the two on regret and on bias. Computed under uniform play, the predicted relative error is vacuous at every setting tried, $K = 40$ through $K = 4000$, while the counted one reads 0.0244 against a budget of 0.0892 at $K = 1000$ and $T = 10^5$.

Interpretation. A prediction pools every action’s history while a count uses one block, which is why counting is noisier, and yet the counted certificate is the better scaled of the two. That setting is the first in which any certificate for ρ_t clears at all.

Limitation. The window that buys it is 53 per cent of the horizon, and the frozen-block regret at that window, 1.2×10^7 , exceeds the 2.6×10^4 the pooling was meant to improve, so what clears is the certificate and not a usable algorithm.

Does the effective dimension, rather than $|\mathcal{S}|$, govern the gain? Holding $|\mathcal{S}| = 8$, $K = 40$, $T = 5000$, $d = 20$ and 24 seeds, and varying only the overlap between the rows of P , the measured mean of v_t moves while $|\mathcal{S}|$ does not:

row overlap	mean v_t	state-pooled	action-level	paired gain
0.00	2.719	392.7	599.3	206.6 ± 20.0
0.25	2.013	361.1	528.1	167.0 ± 21.3
0.50	1.532	306.5	406.5	100.0 ± 10.5
0.75	1.168	199.9	229.2	29.3 ± 3.2
0.95	1.009	49.2	50.8	1.6 ± 0.5

Observed. Both the regret and the advantage over action-level weighting fall monotonically with v_t , from a paired gain of 206.6 at $v_t = 2.719$ to 1.6 at $v_t = 1.009$. Nothing about the raw state count changes across the sweep.

Interpretation. $|\mathcal{S}|$ is constant and cannot explain the pattern, while v_t moves with it, which is the comparison Theorem 1 predicts.

Limitation. The two estimators do not become the same object at $v_t \approx 1$. Rather the rows of P agree, so every action induces the same loss and both regrets collapse, making that endpoint uninformative rather than favorable. One parameter of one map family was varied, so the sweep is consistent with the effective dimension governing the gain rather than establishing it.

Is there a bias-variance trade in the representation? With 16 raw states drawn from four latent clusters, $K = 40$, $T = 6000$, $d = 20$ and 24 seeds, running the algorithm on nested groupings of size 1, 2, 4, 8, 16 gives, for the analyzed $m = d + 1$ variant, regret 1590, 831, 754, 920, 1080 at within-cluster heterogeneity 0, with mean v_t rising 1.000, 1.122, 1.435, 1.663, 2.089 and $\delta_t(g)$ falling 0.700, 0.275, 0, 0, 0.

Observed. The minimum sits at four groups, the number of latent clusters, and stays there at heterogeneity 0.12, where regret reads 1657, 885, 787, 960, 1113. Realized regret stayed below the bound of Theorem 2 in every cell. The bound’s own optimum agrees at heterogeneity 0, where it reads 9364, 4321, 1155, 1243, 1394 and also selects four groups, but not at 0.12, where it reads 11128, 6073, 3430, 3231, 1385 and selects the finest representation.

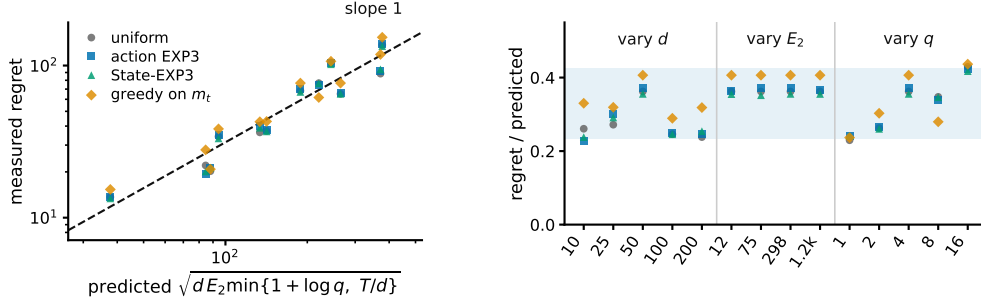


Figure 3: Study 8, the drift channel. **Left**, measured regret against the scale Theorem 3 predicts, over twelve cells in which d varies 20-fold, E_2 100-fold and q 16-fold. The dashed line has slope one. **Right**, the same points divided by that scale, grouped by which parameter the cell sweeps, with the shaded band spanning the observed range. STATE-EXP3 appears here only as one more algorithm on the hard family, where it has no advantage.

701 Interpretation. A trade-off in the representation exists and is located at the latent structure, which is
 702 what Theorem 2 describes qualitatively.

703 Limitation. The bias term $2 \sum_t \delta_t(g)$ is the conservative half of the trade, so the theorem certifies
 704 a trade-off whose location it does not predict once groups are heterogeneous, and the latent cluster
 705 count was known to the experimenter and not to any algorithm.

706 Separately, on the construction of Theorem 3 with $h = d$ the mean pseudo-regret over 400 seeds is
 707 44.1 against a predicted fluctuation of 40.5, and of the five policies compared there the four without
 708 access to the block signs attain expected loss within 0.05% of $T/2$, as Lemma 14 predicts, while the
 709 one given them attains pseudo-regret -954 . The programs producing every number reported here,
 710 together with the stored outputs each figure reads, are at [https://github.com/melikabaghi/](https://github.com/melikabaghi/state-exp3)
 711 state-exp3.

712 **Does the drift bound describe the right scaling?** The construction of Theorem 3 at $K = 40$,
 713 $T = 8000$, $h = d$ and 12 paired seeds, sweeping d from 10 to 200, the amplitude ε from 0.02 to
 714 0.20 so that E_2 runs from 12 to 1193, and the drifting coordinates q from 1 to 16. Four policies are
 715 run, uniform, action-level EXP3, STATE-EXP3, and the greedy policy on the exact stale vector m_t ,
 716 the last because Theorem 3 claims to survive it. E_2 is measured on the realized instance rather than
 717 predicted, so the normalization uses the quantity the theorem uses.

718 Observed. The predicted scale $\sqrt{d E_2 \min\{1 + \log q, T/d\}}$ ranges from 38 to 377 across the grid
 719 and the measured regret tracks it (Figure 3). Normalized, the four policies read 0.230 to 0.429,
 720 0.228 to 0.424, 0.238 to 0.417 and 0.237 to 0.437, a spread below 1.9 against a tenfold range in
 721 the predictor. The oracle-assisted policy has the largest mean ratio, 0.345 against 0.311, 0.314 and
 722 0.310 for the three that never see m_t .

723 Interpretation. The three arguments enter the predicted scale differently and the collapse holds in all
 724 three sweeps, so it is the form of the bound rather than one fitted constant that tracks the data. That
 725 greedy play on the exact m_t sits at the top of the band rather than below it is the concrete content of
 726 the claim that the bound survives the oracle.

727 Limitation. Every cell has $T/d > 1 + \log q$, so only the $1 + \log q$ branch of the minimum is
 728 exercised and the T/d saturation branch, which needs d comparable to T , is not probed. A lower
 729 bound quantifies over all algorithms and four policies cannot establish that. What these runs check
 730 is the scaling of the construction.

731 **Does Theorem 1 describe the right dependence on T , d and $\bar{v}$?** A $4 \times 3 \times 5$ grid in row overlap,
 732 delay and horizon at $K = 40$, $|\mathcal{S}| = 8$ and five paired seeds, running the analyzed $m = d + 1$ variant
 733 and reporting $R/\sqrt{(d+1)\bar{v}T \log K}$. A raw regret table cannot answer this and neither can a bound
 734 that holds with room to spare, since the question is about dependence rather than level.

735 Observed. Over the full grid the ratio spans 35.9 times, from 0.016 to 0.579. Dropping the near-
 736 degenerate column at $\bar{v} = 1.013$ leaves 7.1 times over 45 cells. At $\bar{v} = 3.656$ and $d = 10$ it moves

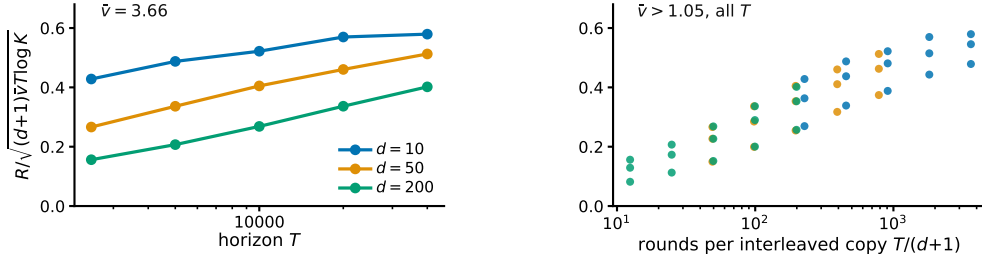


Figure 4: Study 9, the state channel. The plotted ratio would be flat if Theorem 1 captured the dependence on all three of T , d and $\bar{v}$. **Left**, horizon sweep at the largest effective dimension, one line per delay. **Right**, every cell with $\bar{v} > 1.05$ against rounds per interleaved copy, where the three delays fall on one curve.

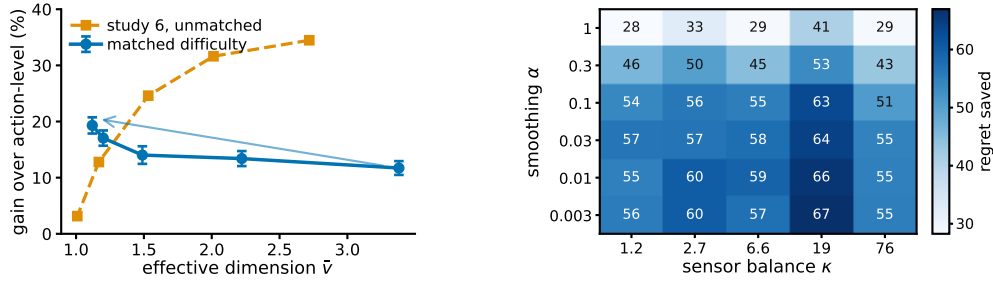


Figure 5: **Left**, study 10. The same overlap sweep with and without holding the task difficulty fixed. Matching the uniform-play regret reverses the trend, so the decline study 6 reports is the confound and not the mechanism. Bars are one standard error of the paired difference. **Right**, study 11, regret saved against action-level weighting over the smoothing pseudocount and the sensor balance, 10 paired seeds per cell.

only 1.35 times across a sixteenfold range of T , and at $d = 200$ it moves 2.57 times. Against rounds per interleaved copy $T/(d+1)$ the three delays fall on one increasing curve (Figure 4).

Interpretation. The $\sqrt{T}$ dependence is well described wherever each copy receives enough rounds, and what is left is carried by the interleaving rather than by T or $\bar{v}$. At $d = 200$ and $T = 2500$ each of the 201 copies sees 12 rounds, so that cell is nowhere near the regime the bound describes. Study 1 reaches the same conclusion about interleaving from the $m = 1$ comparison, and this reaches it from the dependence instead.

Limitation. The degenerate column is degenerate for the reason study 6 gives, since at $\bar{v} \approx 1$ the rows of P agree, every action induces nearly the same loss, and both regrets collapse for a reason the bound cannot see. One natural explanation for the remaining looseness is ruled out rather than confirmed: the realized $\sum_t v_t$ lies within 1.01 to 1.49 of the budget $\bar{v}T$, so the sup over play distributions is not the source.

Does pooling help because the rows overlap, or because overlap made the task easy? Study 6 sweeps the overlap and finds the advantage falling as $\bar{v}$ falls, which read at face value contradicts Theorem 1, whose bound improves as $\bar{v}$ falls. That sweep is confounded, since raising the overlap also flattens $c_t = P\theta_t$ across actions, collapsing the comparator gap so that both regrets vanish together. This study holds $K = 40$, $|\mathcal{S}| = 8$, $T = 10000$, $d = 50$ and the difficulty fixed, and varies only $\bar{v}$. Difficulty is the uniform-play regret $\sum_t \text{mean}_a c_t(a) - \min_a \sum_t c_t(a)$, a property of the instance with no algorithm in it, and the amplitude of θ is bisected in every cell to hit a common target. The map keeps one contrast direction alive at every overlap and θ is aligned with it, which is what lets the gap survive as the rows agree.

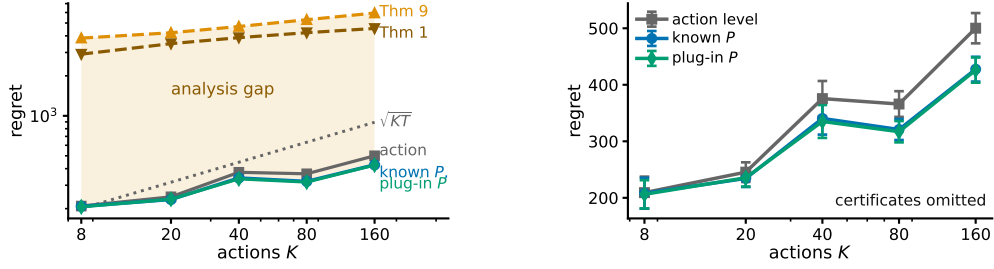


Figure 6: Study 12. **Left**, the three measured arms and the two certificates on one axis, $|\mathcal{S}| = 8$, $T = 5000$, $d = 50$, 12 paired seeds, every measured arm at $m = 1$. The shaded region is what the analysis costs rather than what the method does. **Right**, the same measured arms rescaled with the certificates removed, which is the only way to see that known P and the plug-in coincide. Bars are one standard error.

758 Observed. Across 20 paired seeds the relative gain rises from 11.7 per cent at $\bar{v} = 3.380$ to 19.3 per
 759 cent at $\bar{v} = 1.120$, with absolute gains 85.5 ± 8.9 to 106.2 ± 7.9 and a correlation of -0.929 between
 760 $\bar{v}$ and the gain. Study 6’s unmatched sweep falls from 34.5 to 3.1 per cent over a comparable range
 761 (Figure 5).

762 Interpretation. The two sweeps disagree in sign, and the one that controls difficulty agrees with
 763 the direction Theorem 1 predicts. Overlap is what pooling exploits, and study 6 measures overlap
 764 confounded with an instance becoming trivial. This also explains study 9’s degenerate column
 765 without appealing to it.

766 Limitation. At the two highest overlaps some seeds cannot reach the common target and the realized
 767 difficulty is 5 to 13 per cent below it, so the matching is close rather than exact. Since a smaller
 768 gap should if anything shrink the advantage, the measured rise is conservative in that direction. The
 769 reversal is one construction at one target, and the matched gains are also smaller in magnitude than
 770 the unmatched ones, which is what fixing the difficulty costs.

771 **Is the plug-in fragile in the smoothing constant, and does that depend on coverage?** Study 3
 772 reports one map where the plug-in loses at $\alpha = 1$ and recovers as α falls, which is an anecdote
 773 about a single instance. Here α is swept over 1 to 0.003 against the sensor balance κ of Theorem 9,
 774 controlled from 1.2 to 76 by placing mass ρ on one scarce state and drawing rows around that mean,
 775 at $K = 40$, $|\mathcal{S}| = 8$, $T = 8000$, $d = 50$ and 10 paired seeds.

776 Observed. No cell loses to action-level weighting, the gain running from 28.3 to 66.9 over the 30
 777 cells. The gain roughly doubles as α falls from 1 to 0.03 and is flat below that, while moving along
 778 κ at fixed α changes it far less, the worst-to-best plug-in regret within a κ column being 1.07. At the
 779 better α the plug-in is indistinguishable from known P , the paired difference clearing two standard
 780 errors in one of the five columns and changing sign in another.

781 Interpretation. Of the two parameters the analysis names, the one that matters here is the one the
 782 analysis treats as free. A Laplace default costs about half the available gain at every coverage
 783 level tried, and coverage imbalance over a sixtyfold range costs much less than that. Taken with
 784 study 2 this is further evidence that the difficulty in the unknown- P setting is certification rather
 785 than estimation.

786 Limitation. This is one map family with κ varied through a single scarce state, so it does not rule
 787 out a construction in which κ is the binding parameter, and Theorem 9 still carries κ for a worst case
 788 this family need not contain. The flatness below $\alpha = 0.03$ says the leading behavior stops moving,
 789 not that any fixed α is safe at every horizon.

790 **How far is the method from its certificate, and the certificate from useful?** Study 2 reports these
 791 numbers in separate columns, leaving the reader to assemble them. Here action-level weighting,
 792 STATE-EXP3 with P given, and the same estimator with $\hat{P}_t$ updated every round are measured on
 793 the same paired seeds across K from 8 to 160, and drawn beside the two certificates. Theorem 1 is
 794 the certificate for known P and analyzes $m = d + 1$, while Theorem 9 is the warm-start rate and

analyzes an algorithm that freezes $\hat{P}$ and restarts, so both are plotted to show the size of a gap and not as like for like.

Observed. The plug-in stays within 1.6 per cent of known P at every K , the two curves being visually indistinguishable, while action-level weighting costs 1.4 to 17.4 per cent more than the plug-in as K rises twentyfold. Theorem 1 sits 10.7 to 14.8 times above the measured plug-in and Theorem 9 sits 14.0 to 18.7 times above it. The warm-start certificate also stays above $\sqrt{KT}$ at every K tried, by factors of 19.2 down to 6.6.

Interpretation. The vertical gap is the analysis, not the method. Estimating P online is close to free on this family, so the order of magnitude between the measured arms and either certificate is conservatism in the proof. That the warm-start rate never drops below $\sqrt{KT}$ here is the concrete form of the statement in Section 4 that it does not certify the channel it analyzes.

Limitation. The measured arms run $m = 1$ and Theorem 1 analyzes $m = d + 1$, and the plug-in is not the warm-start algorithm Theorem 9 analyzes, so neither vertical gap is a statement about a bound being loose for the algorithm it covers. A conservative bound and a well-behaved algorithm are consistent with a third possibility, that the family is easy.

Can the coarsening be chosen from data? Theorem 2 takes the representation as given and study 7 uses the latent cluster count, which no algorithm knows. This study supplies a rule and measures what it costs, with no theorem behind it. A prefix of 1200 rounds runs the identity representation and estimates $\hat{\theta}(s)$ by averaging the outcomes seen at each state. Sorting states by $\hat{\theta}$ and merging adjacent pairs greedily gives one candidate at every granularity from 16 down to 1, each scored by the quantity Theorem 2 bounds, $\sqrt{2(d+1)V(g)\log K} + 2\sum_t \hat{\delta}_t(g)$, and the argmin runs the remaining rounds. Against it are the identity, the true clusters, and the best candidate on the ladder scored in hindsight, over 24 paired seeds.

Observed. The rule fails. It selects 15 groups at heterogeneity 0 and 16 at 0.12, against a true cluster count of four, and lands at 912.9 ± 30.1 and 944.7 ± 31.1 where the identity gives 921.0 ± 30.5 and 952.7 ± 32.4 , so it captures under one per cent of the available gain. The true clusters give 843.7 ± 27.9 and 883.0 ± 31.1 , and hindsight over the same ladder gives 833.5 ± 27.8 and 841.5 ± 23.6 .

Interpretation. Hindsight nearly matches the true clusters, so the ladder does contain a good representation and the failure is the criterion rather than the candidate set. Measuring $\hat{\delta}$ at the four-group representation locates it. At heterogeneity 0 the true within-group spread is exactly zero while the prefix estimates it at 0.154, which is sampling noise, and the bound multiplies that by $2(T - T_0)$ to give a phantom bias of 1483 against a real variance saving of 1261. The bias term is the conservative half of Theorem 2, and a noisy plug-in estimate of it is enough to forbid every merge.

Limitation. One selector on one family, and a negative result about this rule is not a negative result about data-driven selection. The obvious repairs, debiasing $\hat{\delta}$ for its sampling noise or scoring candidates on held-out loss instead of on the bound, are not evaluated here. The prefix is also charged to every arm equally, so the comparison understates what a selector that spent less on estimation might do.

Is “many actions resolved by few states” a natural regime? Every study so far draws P from a Dirichlet, which is the assumption under test rather than evidence for it. This one builds an environment to the shape of a recommendation funnel instead. Actions are $K = 200$ catalogue items, states are $|\mathcal{S}| = 6$ ordered engagement depths, and the outcome is a delayed conversion. Three structural facts are imposed because funnels have them, namely that items fall into a modest number of categories sharing an engagement profile, that item quality is heavy-tailed, and that the loss falls with engagement depth and drifts seasonally rather than adversarially. The generated catalogue puts 74 per cent of its mass on no engagement and 1.4 per cent on the deepest.

Observed. The effective dimension is $\bar{v} = 1.97$ against $K = 200$, a hundredfold gap, and it stays between 1.41 and 1.81 as the catalogue grows from 25 to 800 items (Figure 7). Against action-level weighting the state-pooled variant saves 79.3, 63.6 and 38.9 per cent at $d = 10$, 50 and 200, the largest margins anywhere in this appendix. The plug-in matches known P at every delay, the paired difference never reaching two standard errors. The regime condition $\bar{v}(d+1)\log K \leq K + d$ holds at $d = 10$ and fails at $d = 50$ and $d = 200$.

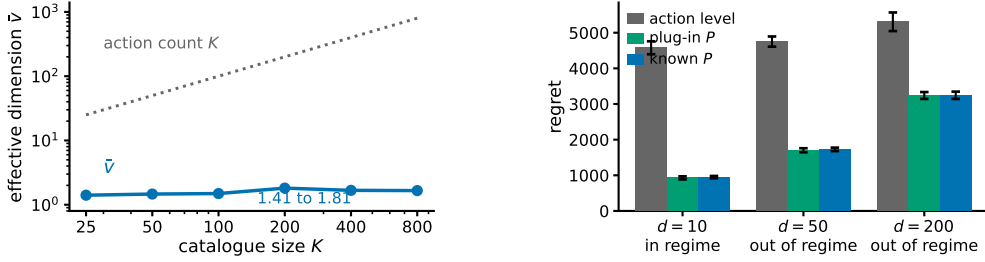


Figure 7: Study 14, a recommendation funnel. **Left**, the effective dimension stays near two while the catalogue grows thirty-two fold, so the gap between K and $\bar{v}$ comes from the catalogue having categories, not from a chosen K . **Right**, the three arms at $T = 20000$ and 8 paired seeds, labelled by whether the regime condition of Section 4 holds. Bars are one standard error. No real data is used anywhere in this study.

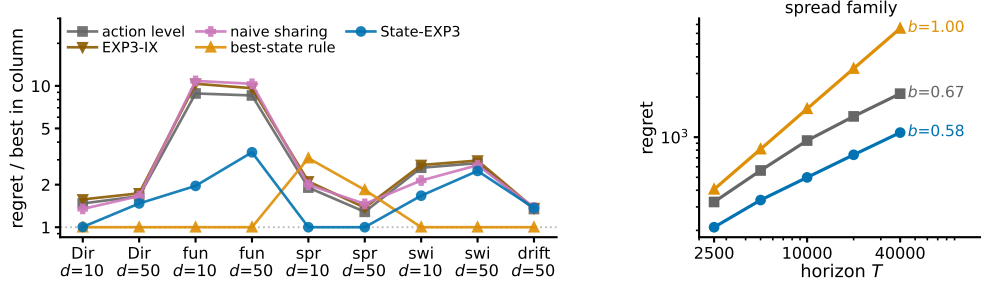


Figure 8: Study 15. **Left**, regret relative to the best arm in each family, so 1 is that column’s winner, over five policies, nine settings and 10 paired seeds. The best-state rule leads seven columns and STATE-EXP3 leads two. **Right**, growth in T on the one family built so the rule’s assumption is false, with fitted exponents. The left panel is a worst case over families chosen here; the right panel is why that is not a worst case over the model.

847 Interpretation. The separation between K and $\bar{v}$ arrives from catalogue structure rather than from
848 a tuned parameter, and it does not close as the catalogue grows, which is the claim the Dirichlet
849 families cannot make. That the condition Theorem 1 needs fails at the delays a funnel has, while
850 the practical variant still saves most of the regret, is the same message as study 1 in the setting that
851 motivates the paper. It is evidence for Conjecture 4 rather than for the theorem.

852 Limitation. *No real data is used.* There is no interaction log in this environment, nothing here
853 is fitted to one, and this is not a semi-synthetic benchmark in the usual sense of that phrase. The
854 structure is imposed by hand from qualitative properties, so it shows that the regime is describable
855 and self-consistent, not that any deployed system occupies it. The arms run $m = 1$, which carries no
856 guarantee, and the environment has no adversarial component, so it exercises the pooling channel
857 and not the drift channel.

858 **Which assumption produces which benefit?** Comparing against action-level weighting alone
859 leaves open whether the gain comes from the state structure or from an implementation detail. Five
860 policies are run on the same paired seeds over nine settings. Besides action-level weighting and
861 STATE-EXP3 they are EXP3-IX, the usual robustification of an importance weight; *naive sharing*,
862 which credits $P(S_t | a)X_t / \bar{p}(S_t)$ using the unconditional state distribution and so isolates the value
863 of the denominator $q_t = x_t^\top P$ from the value of sharing at all; and a *best-state rule*, which estimates
864 $\hat{\theta}(s)$ online, takes the best state, and plays the action most likely to reach it. Two settings are added
865 to the earlier families, one where the best action mixes over several good states rather than maximiz-
866 ing the single best one, and one that is obviously nonstationary. Running a bandit over states and

mapping the values back through P is not a separate baseline, since $\sum_s P(s | a) \mathbf{1}\{S_t = s\} X_t / q_t(s)$ equals $P(S_t | a) X_t / q_t(S_t)$ identically, so that method is STATE-EXP3.

Observed. STATE-EXP3 does not win most columns. The best-state rule leads seven of the nine and STATE-EXP3 leads two, and over these nine settings the rule is also better on the median, 1.00 against 1.47 times the column winner, and on the worst case, 3.08 against 3.39 (Figure 8). EXP3-IX is 1.5 to 17.1 per cent worse than plain action-level weighting everywhere, and naive sharing ranges from 18.7 per cent better to 22.7 per cent worse than it. On the spread family the ordering inverts, STATE-EXP3 leading by 47.6 and 22.2 per cent while the best-state rule falls 61.5 and 43.0 per cent behind action-level weighting. Sweeping the horizon there, the rule’s regret grows with fitted exponent 1.00 and STATE-EXP3’s with 0.58.

Interpretation. The best-state rule buys its lead with an assumption, that the best action is the one most likely to reach the single best state. Where that holds it is very strong, and in the funnel it holds in 10 of 10 seeds because engagement depth is ordered, which is why it leads there by a factor of two. Where it fails the rule commits to a fixed action and pays regret linear in T , which no exploration rate repairs, and the exponent of 1.00 is that happening. The comparison therefore separates the assumptions rather than ranking the methods. STATE-EXP3 assumes nothing about the shape of θ . Across these nine selected settings its largest regret ratio to the column winner is 3.39, and the rule’s suite maximum is slightly smaller at 3.08, but the rule’s fitted growth on the spread family is linear. Action-level weighting also carries a sublinear guarantee, so what separates the arms here is the constant rather than the rate. STATE-EXP3’s paired standard errors are the smallest of the five at 5.8 per cent of the mean against 23.6 for the rule. That naive sharing can be worse than no sharing is the separate point that the denominator, not the sharing, is what Proposition 6 buys.

Limitation. Nine settings chosen here are not a worst case over the model, and the median and worst-case columns above should be read as describing this set only. A practitioner whose problem has the ordered structure of the funnel should use the best-state rule, and this appendix does not identify when that structure can be verified in advance. On the drift family no arm’s paired difference from action-level weighting reaches two standard errors, the largest being the best-state rule’s $+24.6 \pm 12.4$, so that column is reported as an absence of signal rather than a result.

Taken together the fifteen studies cover both channels. Studies one, six, seven, nine, ten, fourteen and fifteen probe the state channel. They are consistent with the effective dimension, rather than $|\mathcal{S}|$ or K , governing what pooling buys, with the qualification that study ten had to control for task difficulty before study six’s sweep pointed the way Theorem 1 does, and study nine shows where the theorem stops describing the dependence and attributes that to the interleaving. Studies two to five, eleven and twelve probe what happens when the action-to-state matrix must be estimated, and locate the difficulty in certifying the estimate rather than in forming it, study eleven adding that the smoothing constant matters more than the coverage imbalance the analysis carries and study twelve putting the measured arms and their certificates on one axis. Study thirteen closes the representation thread by showing that the bound of Theorem 2 is too conservative to serve as its own selection rule. Study fourteen is the only one whose action-to-state matrix is not drawn from a Dirichlet, and it reports the largest margins here, though on a structure imposed by hand rather than measured. Study fifteen widens the baseline set and finds STATE-EXP3 beaten in seven of nine settings by a heuristic that assumes the best action is the one reaching the single best state, which holds on several of these families and fails with linear regret when it does not. Study eight is the drift channel, and its collapse across d , E_2 and q is the strongest evidence here that Theorem 3 has the right form. No study runs a method that uses both channels, which is the gap Section 7 describes.

J Why the delay does not become additive

Run STATE-EXP3 with $m = 1$, so $x_t \propto \exp(-\eta L_t)$ with $L_t = \sum_{r \leq t-d-1} \hat{c}_r$. Let $\tilde{x}_r$ be the iterate that has also absorbed the outstanding estimates $G_r = \sum_{s=r-d}^{r-1} \hat{c}_s \geq 0$. Two terms appear.

The delay term is $\mathbb{E}[\langle x_r - \tilde{x}_r, \hat{c}_r \rangle]$. Since $\tilde{x}_r(a) \geq x_r(a) e^{-\eta G_r(a)}$ and $1 - e^{-z} \leq z$, it is at most $\eta \mathbb{E}[\sum_a x_r(a) G_r(a) \hat{c}_r(a)] = \eta \sum_{s=r-d}^{r-1} \mathbb{E}[\langle x_r, \hat{c}_s \rangle]$, and x_r depends only on rounds $r - d - 1$ and earlier, hence on rounds before s , so $\mathbb{E}[\langle x_r, \hat{c}_s \rangle] = \mathbb{E}[\langle x_r, c_s \rangle] \leq 1$. This term is $\eta d T$ for any unbiased estimator, which is the additive form.

919 The quadratic term is where it stops. The telescoping scores $\sum_a x_{r+d}(a)\hat{c}_r(a)^2$, with x_{r+d} the iter-
 920 ate that has absorbed rounds up to $r-1$, while Proposition 6 bounds the play-weighted sum, the one
 921 against x_r , by v_r . Carrying out the substitution, $\mathbb{E}[\sum_a x_{r+d}(a)\hat{c}_r(a)^2 \mid \cdot] \leq \sum_s q_{r+d}(s)/q_r(s) \leq$
 922 $|\mathcal{S}|/Z_r$ for $Z_r = \mathbb{E}_{a \sim x_r}[e^{-\eta G_r(a)}]$, and $Z_r \geq e^{-\eta \langle x_r, G_r \rangle}$, so the swap costs an exponential moment
 923 of $\langle x_r, G_r \rangle$.

924 No bound on that moment holds uniformly over state distributions. For this estimator $\langle x_r, \hat{c}_s \rangle =$
 925 $q_r(S_s)X_s/q_s(S_s)$, so the quantity is governed by $1/q_s(S_s)$, the reciprocal probability of the state
 926 actually observed. Its mean is $\sum_s q_s(s)/q_s(s) = |\mathcal{S}|$, which is why the delay term is fine. For a
 927 fixed q of full support the exponential moment is finite, $1/q(S)$ being bounded, and what fails is
 928 uniformity. Already at $|\mathcal{S}| = 2$ with $q = (\varepsilon, 1 - \varepsilon)$, $\mathbb{E}[e^{\eta/q(S)}] \geq \varepsilon e^{\eta/\varepsilon} \rightarrow \infty$ as $\varepsilon \rightarrow 0$ while the
 929 mean stays at 2, so no bound in $|\mathcal{S}|$ and η alone is available, and q_s is set by the algorithm's own
 930 past play. Measured at $|\mathcal{S}| = 4$, the mean stays at 4.0 while $\mathbb{E}[e^{0.05/q(S)}]$ runs 1.22, 1.43 and then
 931 1.3×10^{21} as the smallest state probability falls through 0.25, 0.019 and 0.0009.

932 Flooring q restores the moment and costs the product back. Mixing with the uniform action at rate
 933 γ gives $q_t(s) \geq \gamma \bar{p}(s)$, so $\eta \langle x_r, G_r \rangle \leq 1$ needs $\gamma \asymp \eta d \kappa |\mathcal{S}|$ under the balanced-sensor condition
 934 $\bar{p}(s) \geq 1/(\kappa |\mathcal{S}|)$, and the mixing cost γT then reproduces $\tilde{O}(\sqrt{d \kappa |\mathcal{S}| T \log K})$. So the obstruction
 935 is not an artifact of one device. Empirically it does not bite, $1/Z_r$ staying below 1.33 at the tuned η
 936 over $d \in \{10, 50, 200\}$, which is why Conjecture 4 is stated rather than abandoned.

937 **K Stationary sensors**

938 *Observation 17.* The multiplicative term $\sqrt{\min\{|\mathcal{S}|, d\}T}$ is not forced by the known construction,
 939 and is absent at one endpoint of the present model class.

940 The $\Omega(\sqrt{\min\{|\mathcal{S}|, d\}T})$ construction of Esposito et al. (2023, Thm. 5.2) re-splits states across ac-
 941 tions adversarially per block, so its action-to-state map is time-varying and lies outside the model
 942 class of Section 3. At $E_2 = 0$ with stationary P the problem is a stochastic delayed bandit, where
 943 delay enters additively (Joulani et al., 2013). One route to it is therefore unavailable and the term is
 944 absent at one endpoint, but neither a matching upper bound for all $E_2 > 0$ nor a lower bound ruling
 945 it out is established here.